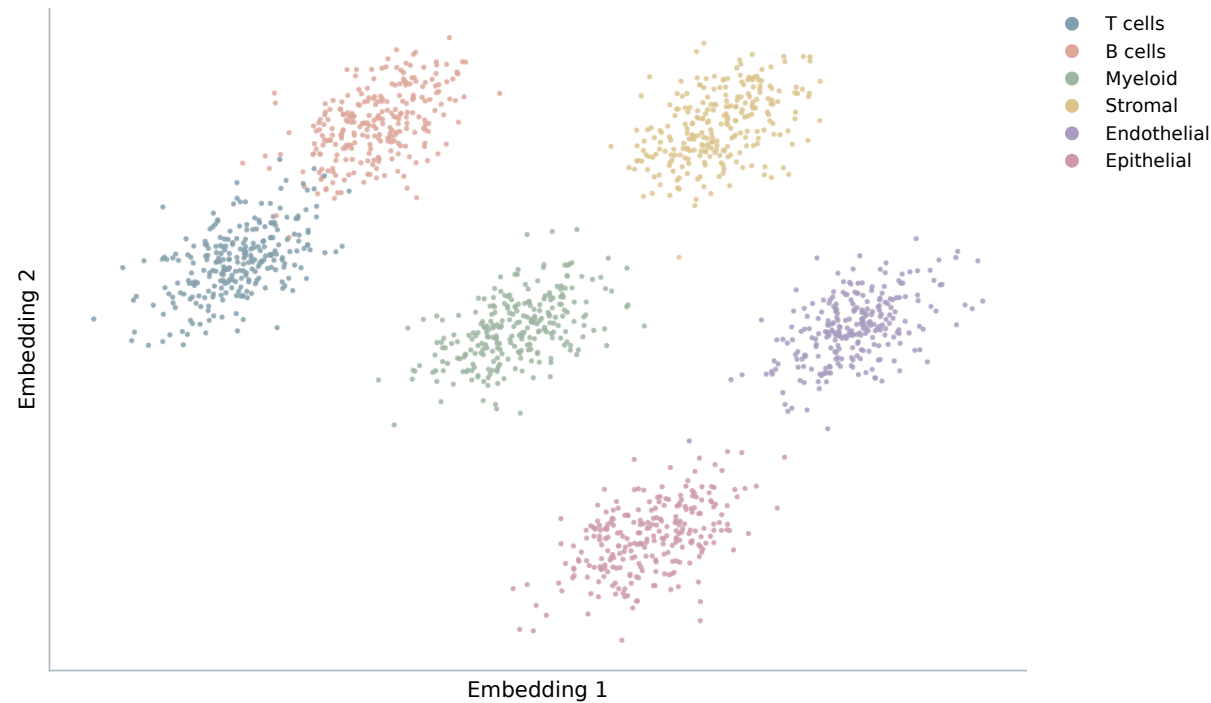
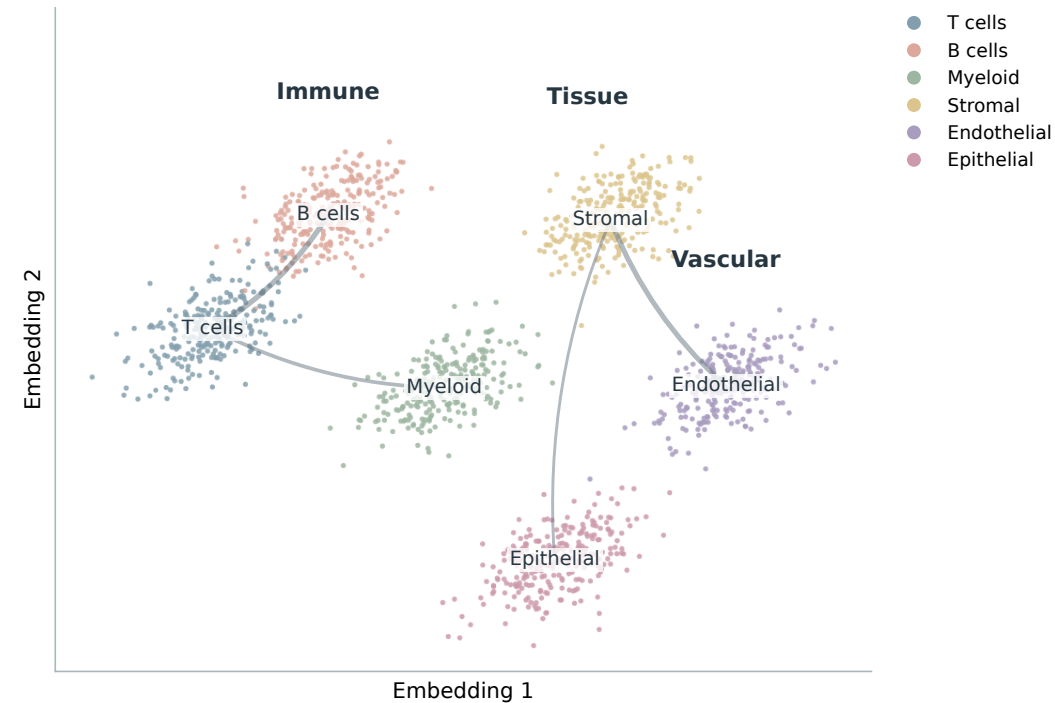


Cell identity and lineage hierarchy



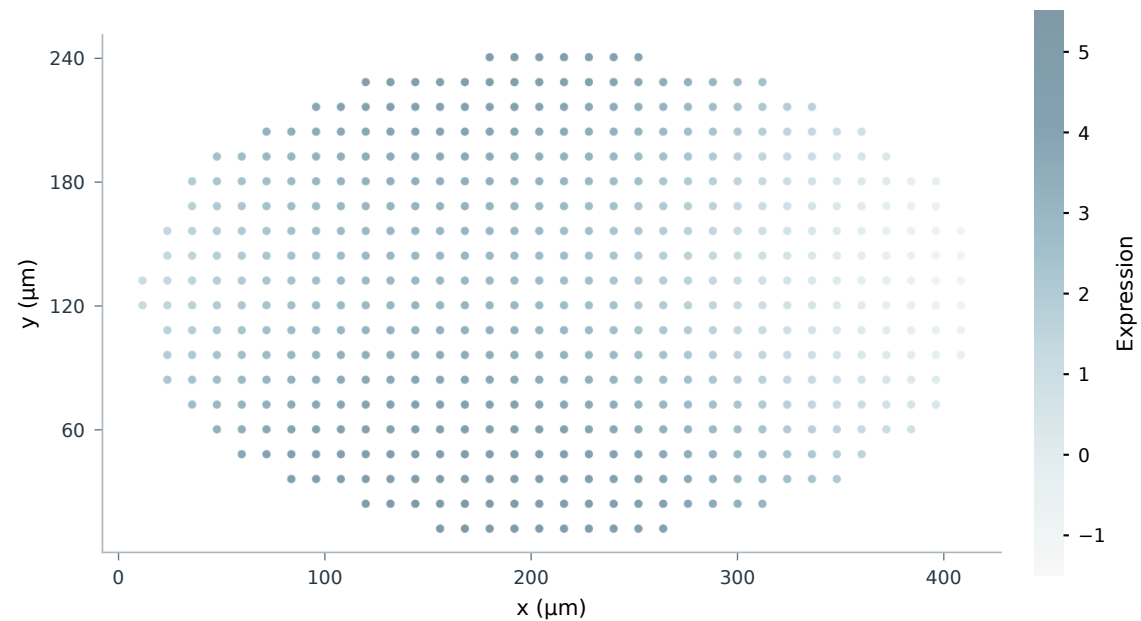
SYNTHETIC EXAMPLE

Cell identity and lineage hierarchy



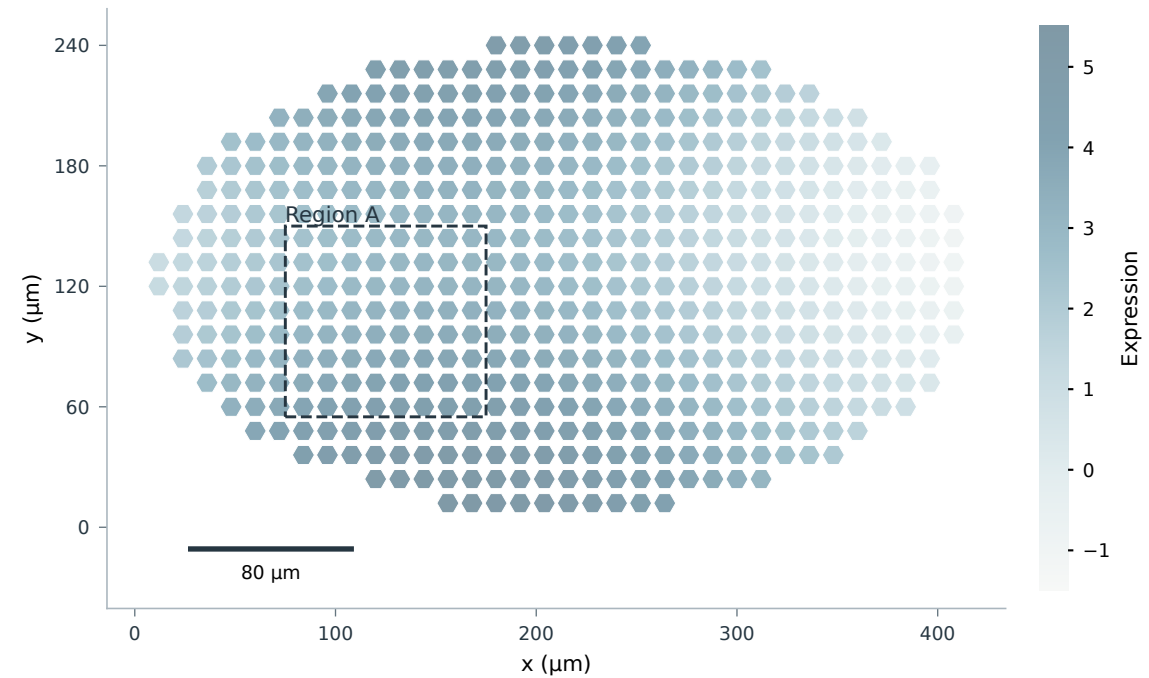
SYNTHETIC EXAMPLE

Spatial expression and tissue compartments



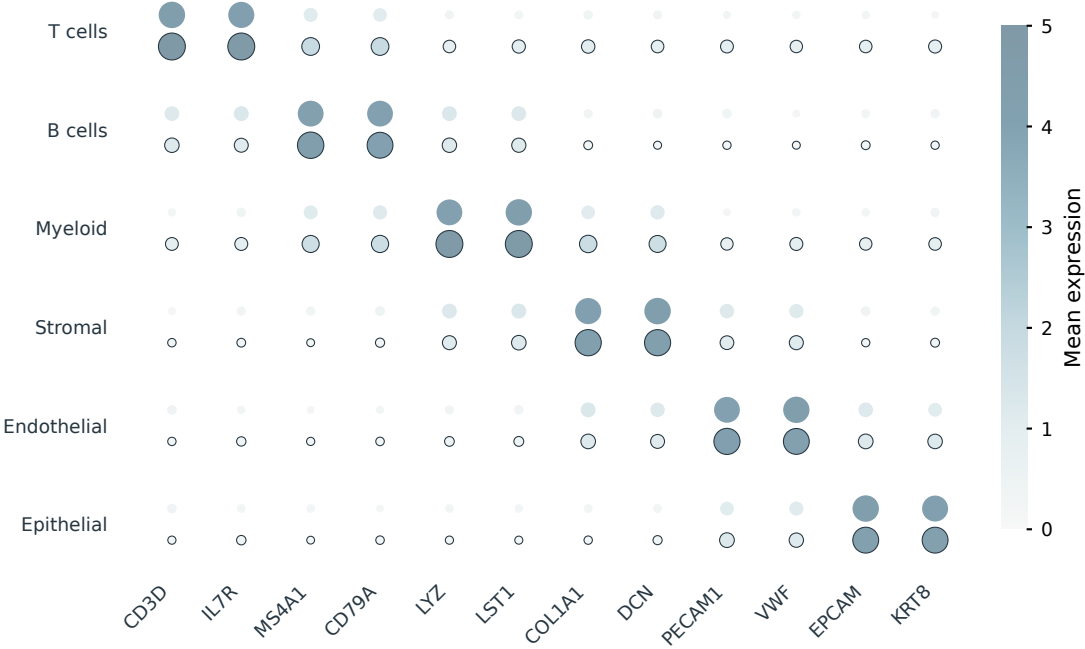
SYNTHETIC EXAMPLE

Spatial expression and tissue compartments



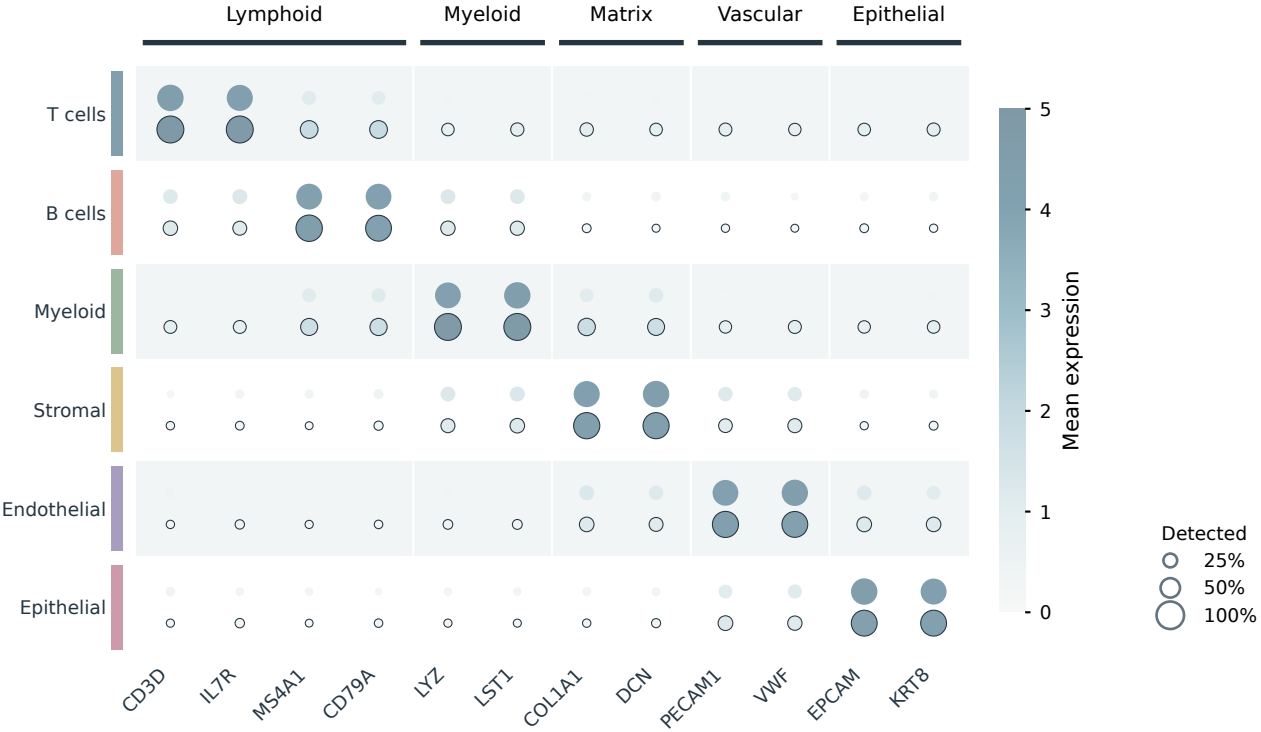
SYNTHETIC EXAMPLE

Marker programmes across cell states



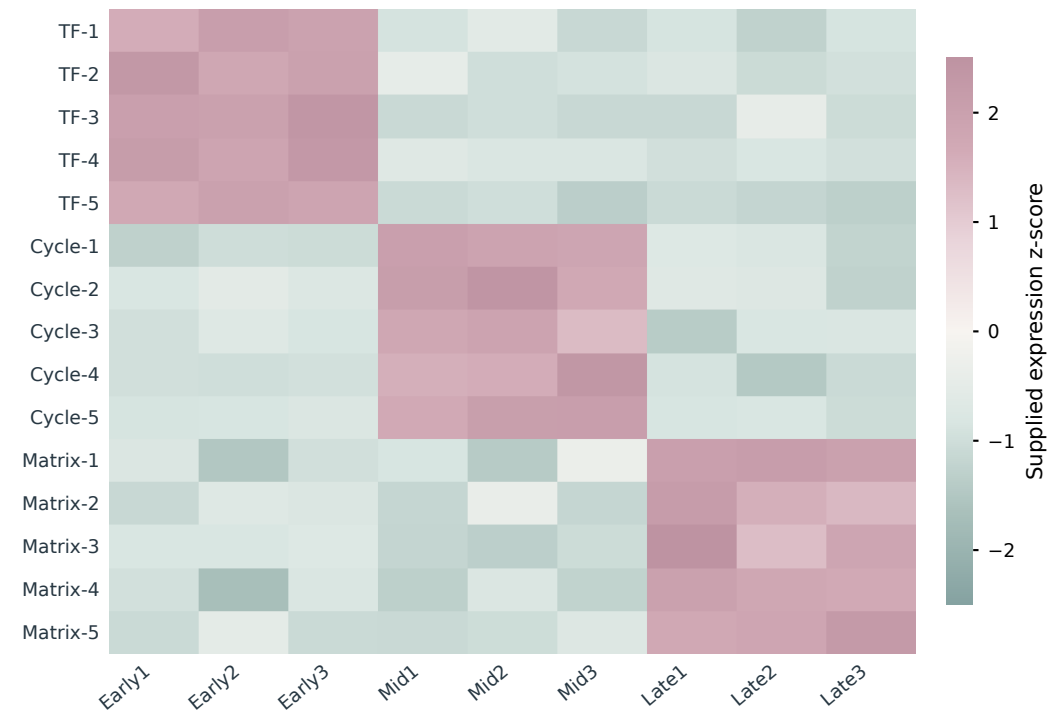
Row order: Baseline / Stimulated
SYNTHETIC EXAMPLE

Marker programmes across cell states



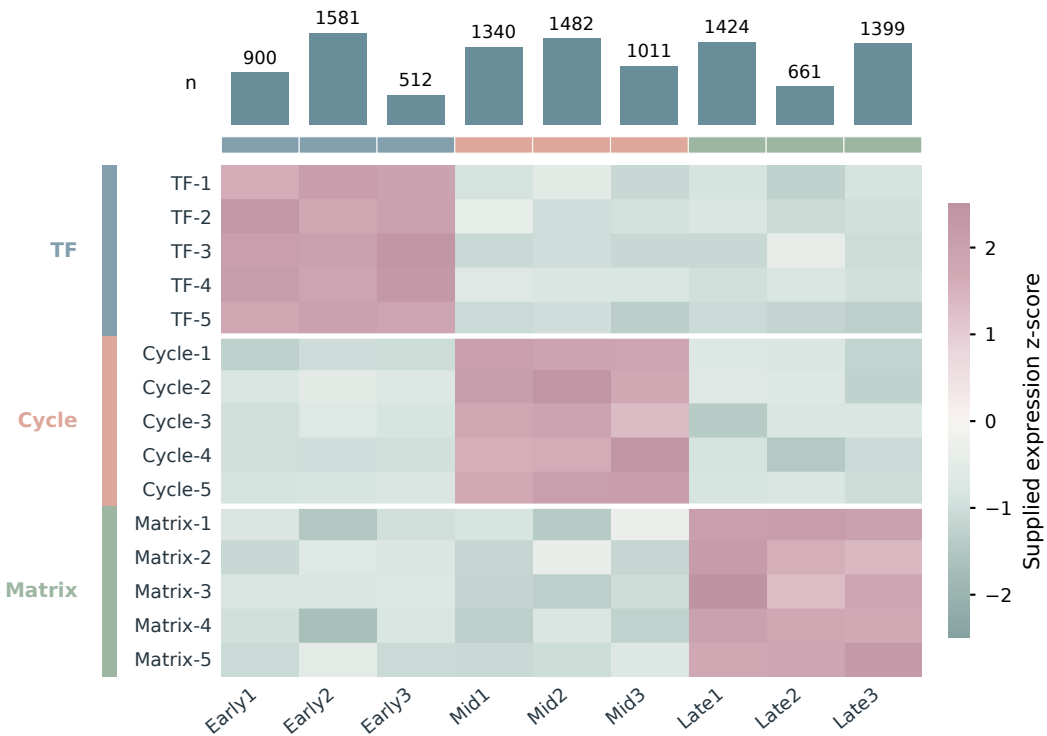
Row order: Baseline / Stimulated
SYNTHETIC EXAMPLE

Expression modules and sample context



SYNTHETIC EXAMPLE

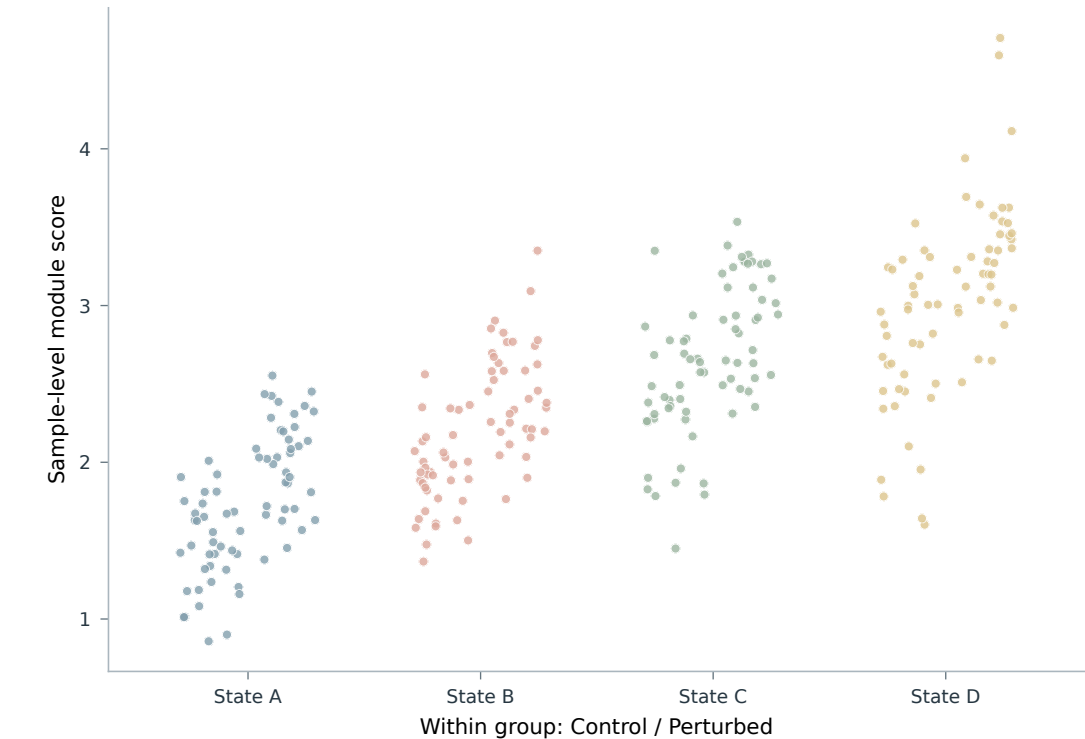
Expression modules and sample context



SYNTHETIC EXAMPLE

Condition
Early Mid Late

Distribution across biological samples



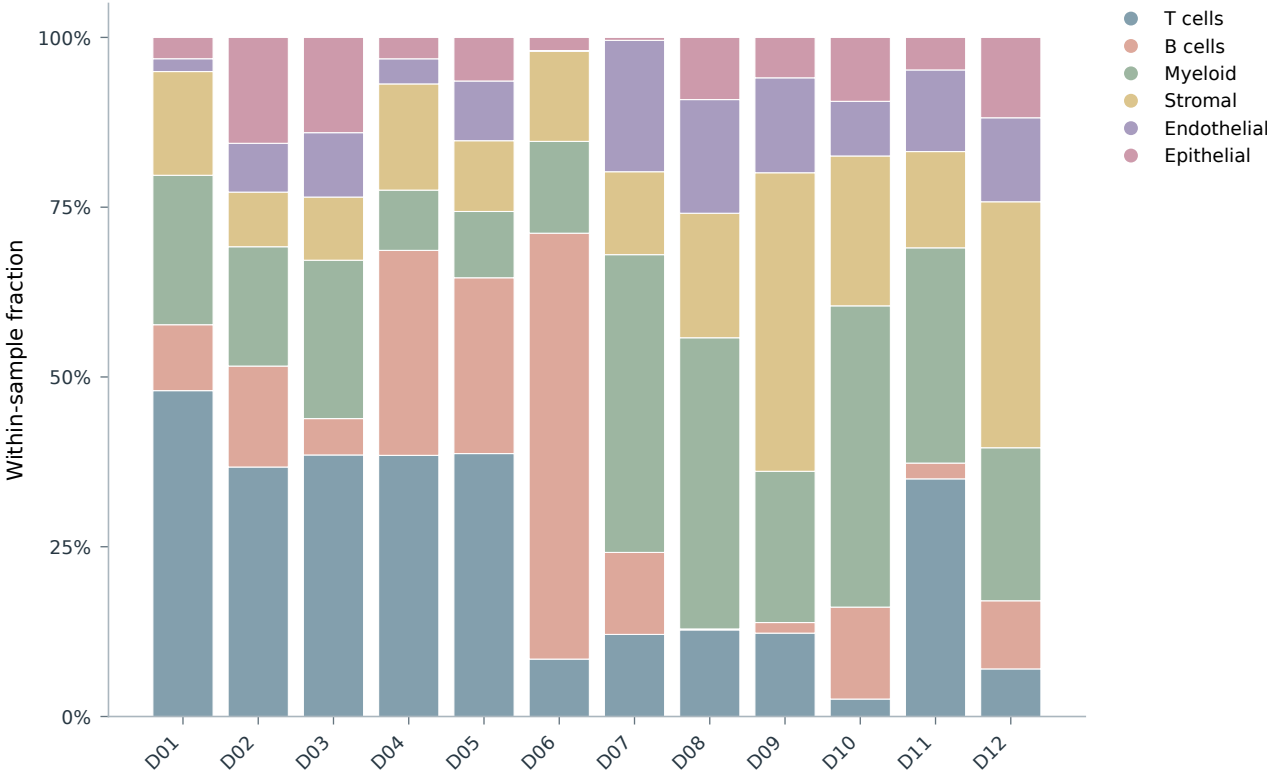
SYNTHETIC EXAMPLE

Distribution across biological samples

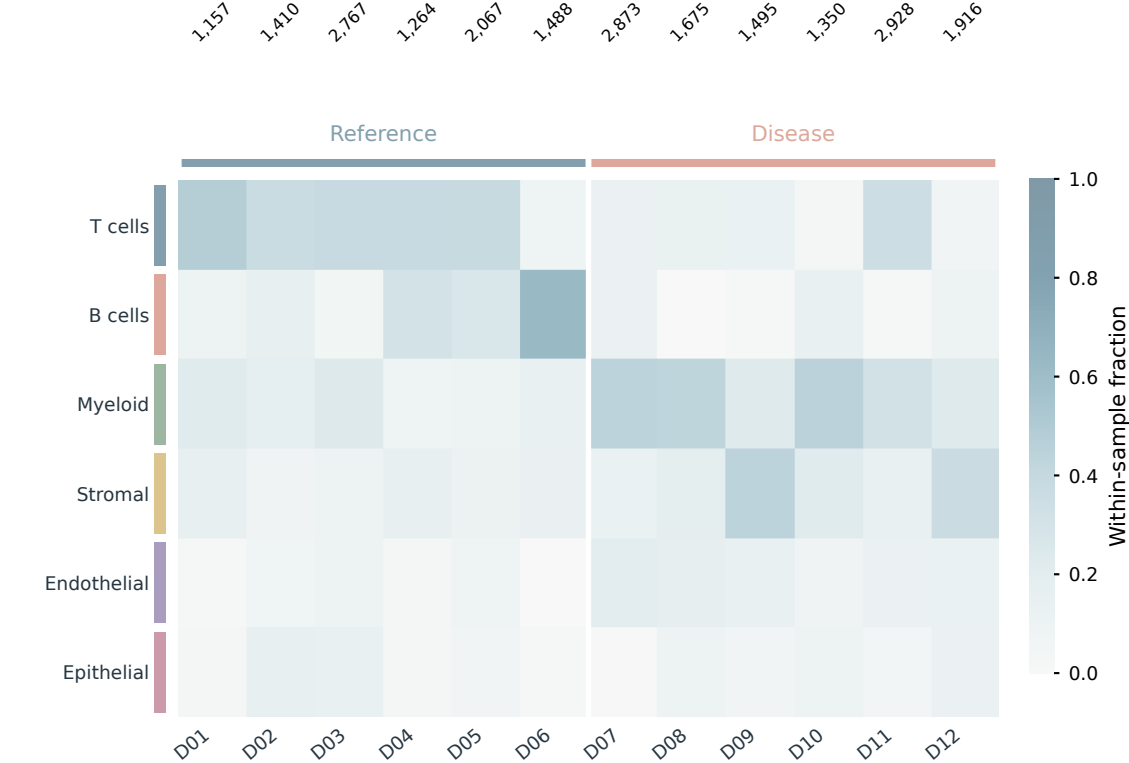


SYNTHETIC EXAMPLE

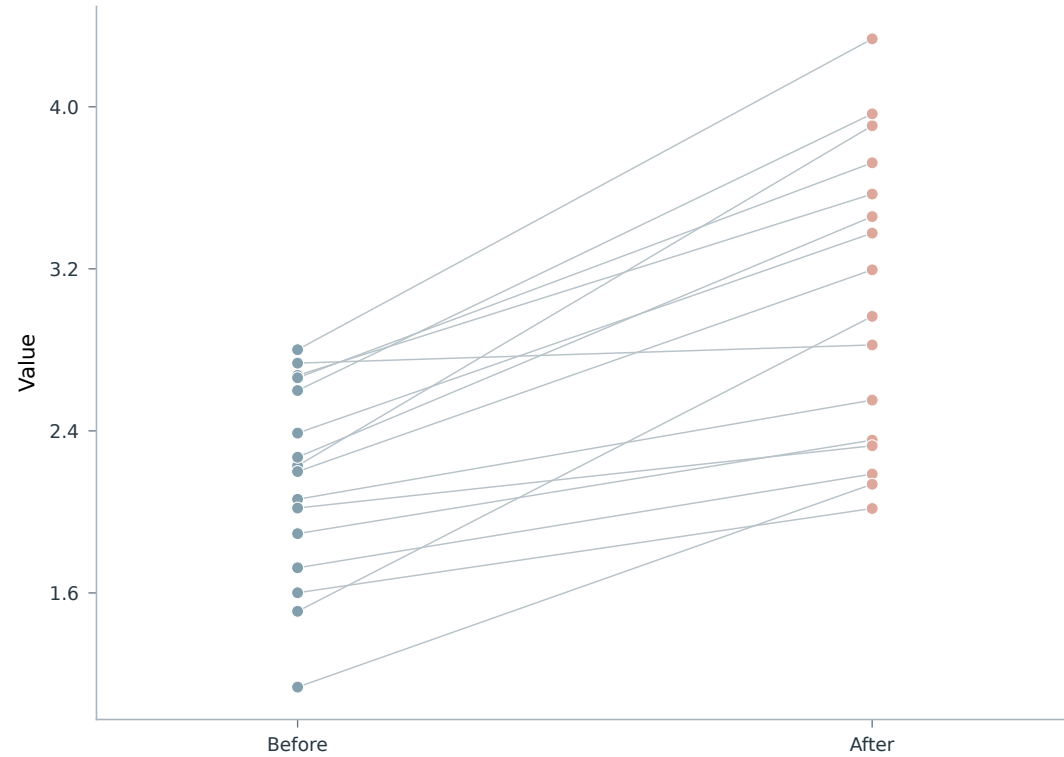
Cell composition across samples



Cell composition across samples

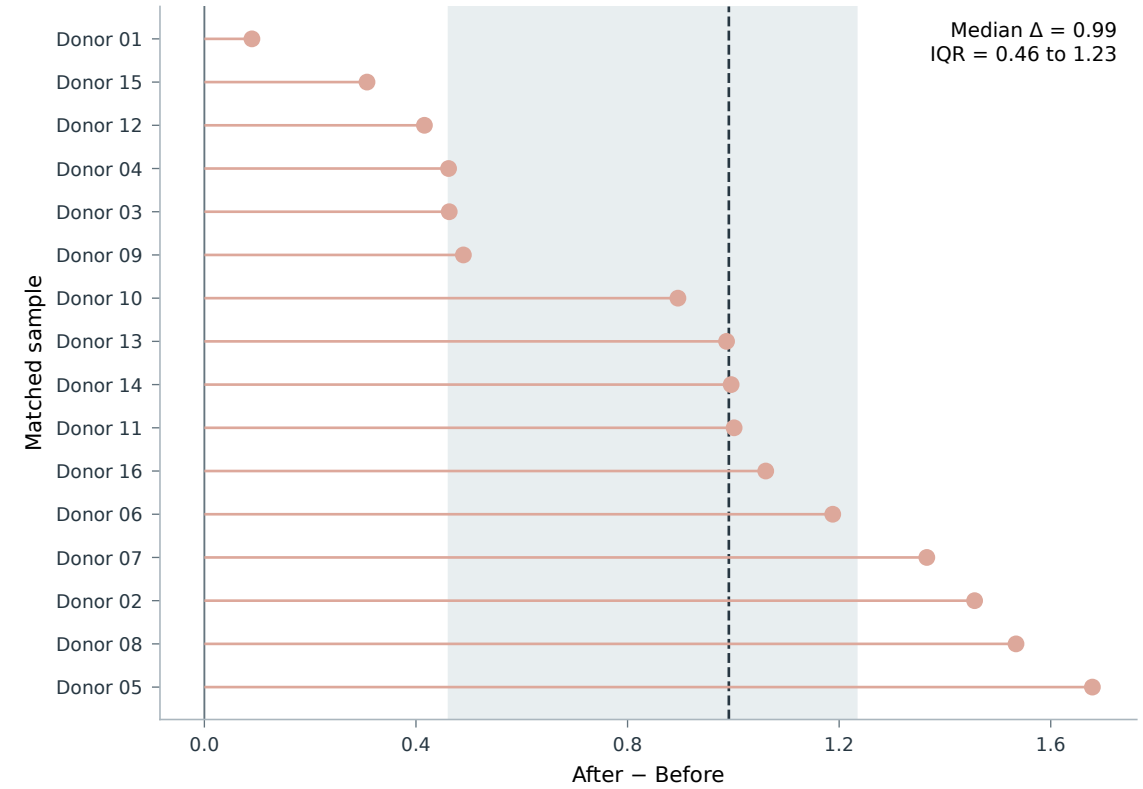


Matched sample responses



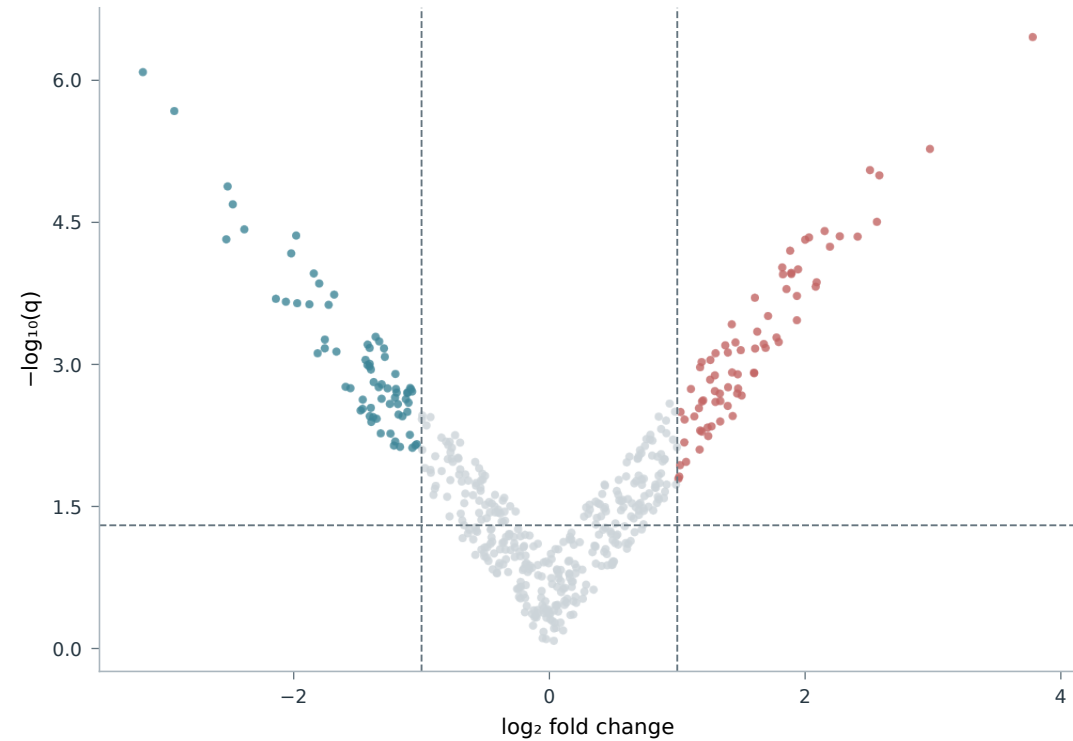
SYNTHETIC EXAMPLE

Matched sample responses



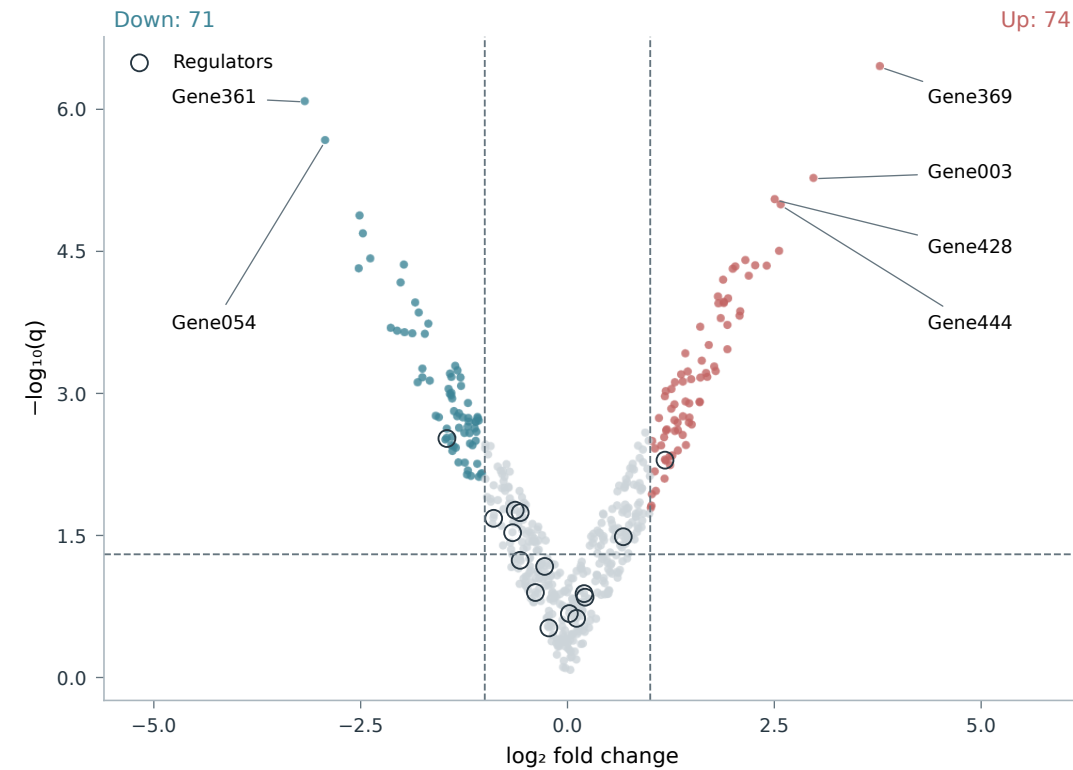
SYNTHETIC EXAMPLE

Differential expression and annotated gene sets



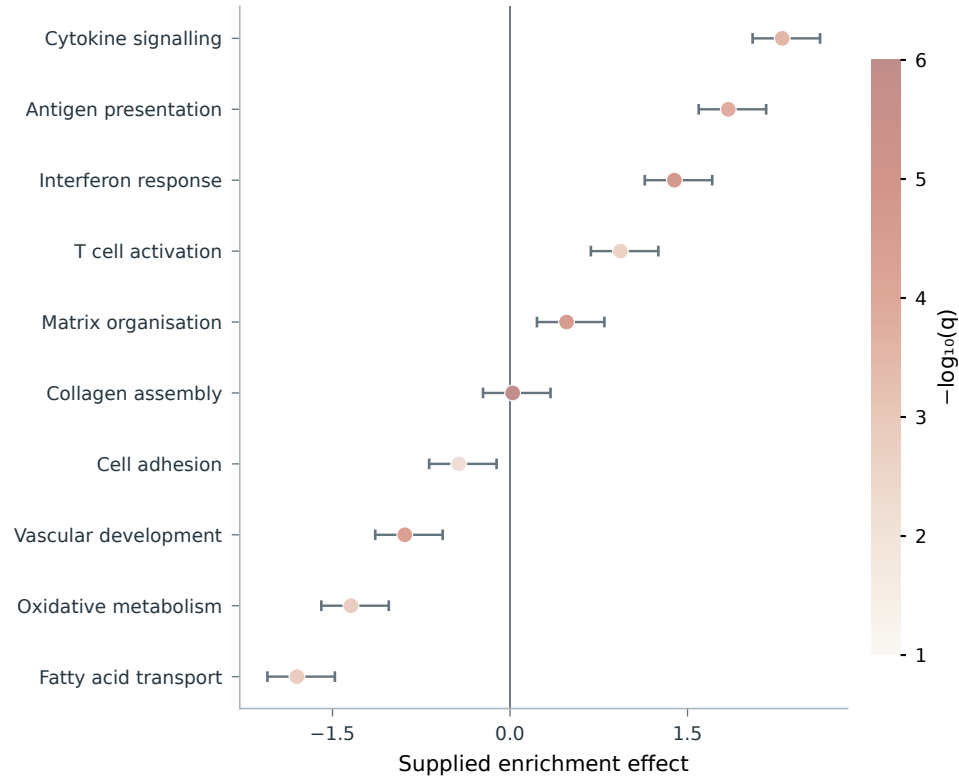
SYNTHETIC EXAMPLE

Differential expression and annotated gene sets



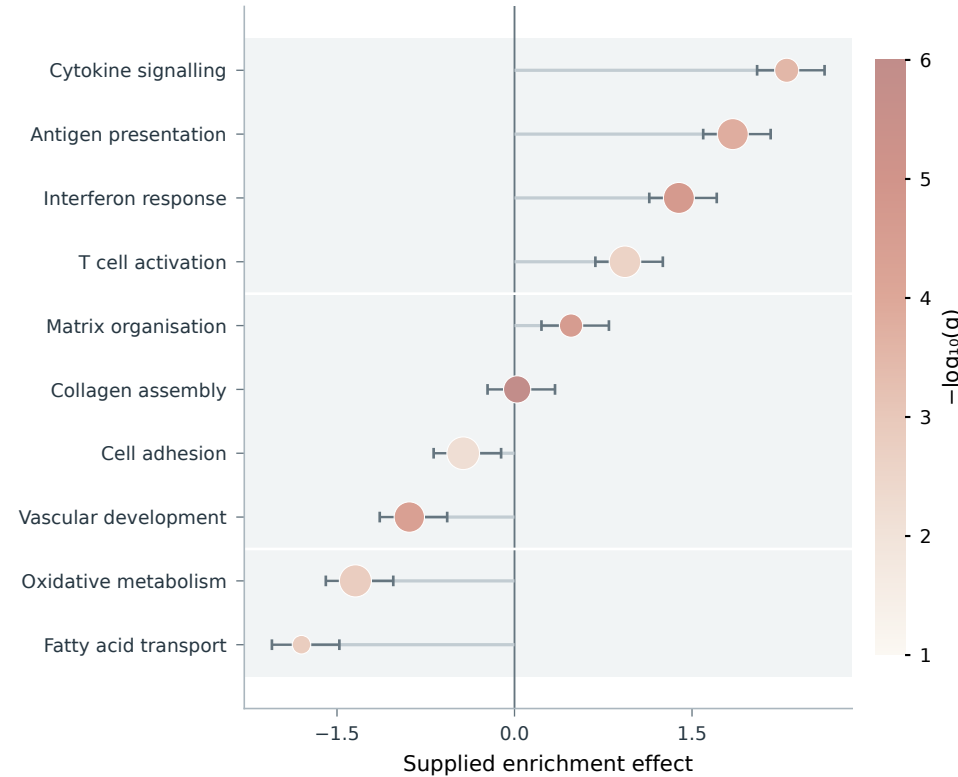
SYNTHETIC EXAMPLE

Pathway effects and supporting evidence



SYNTHETIC EXAMPLE

Pathway effects and supporting evidence

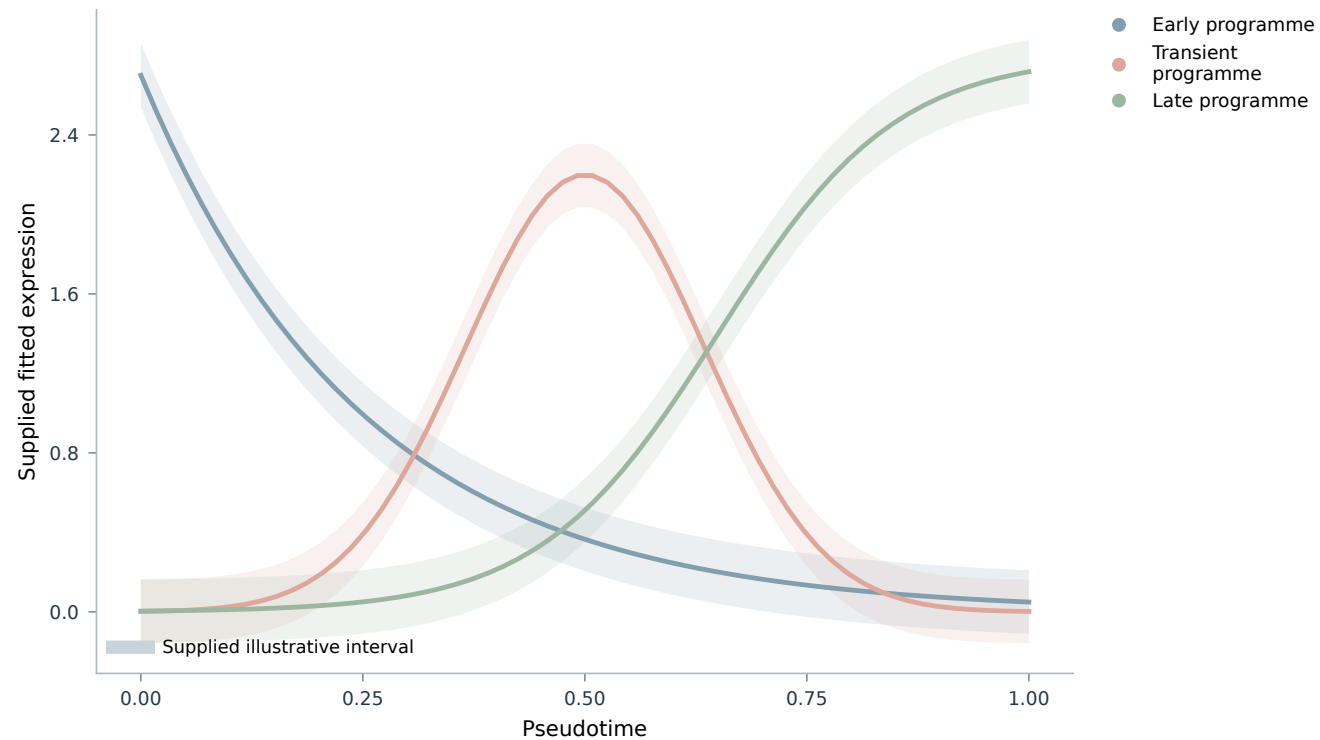


SYNTHETIC EXAMPLE

Genes

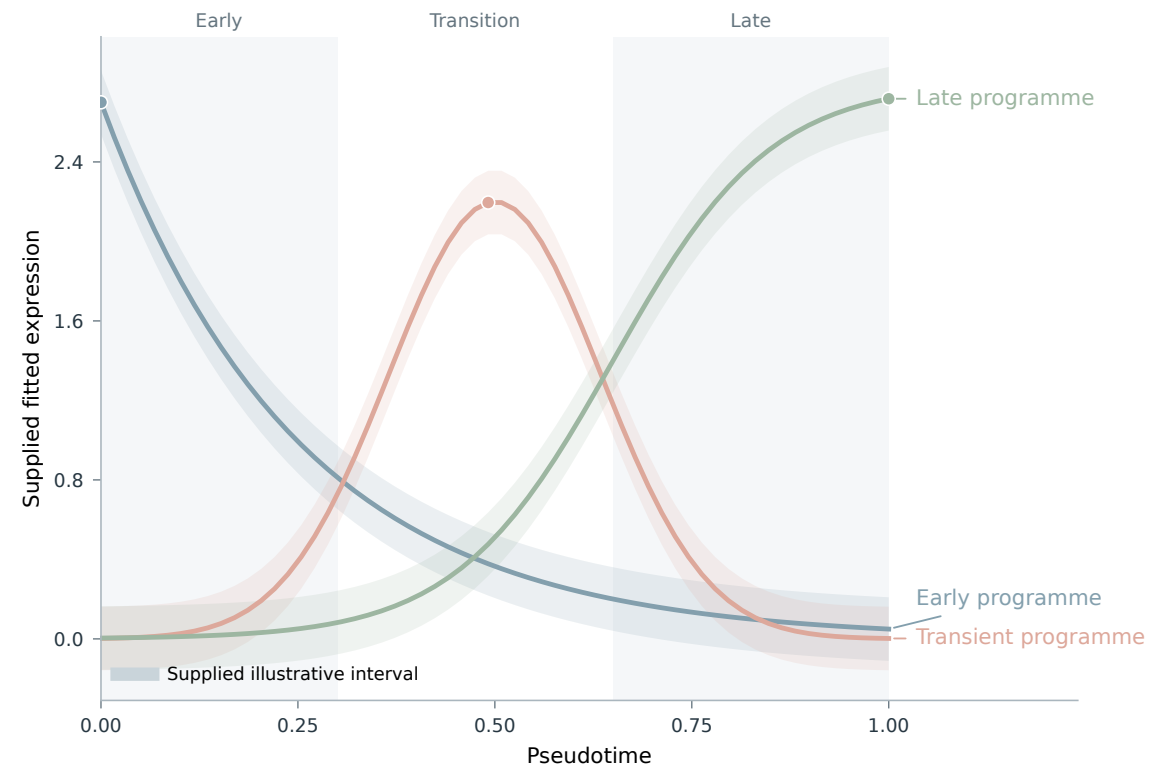
- 34
- 52
- 59

Lineage programmes over time



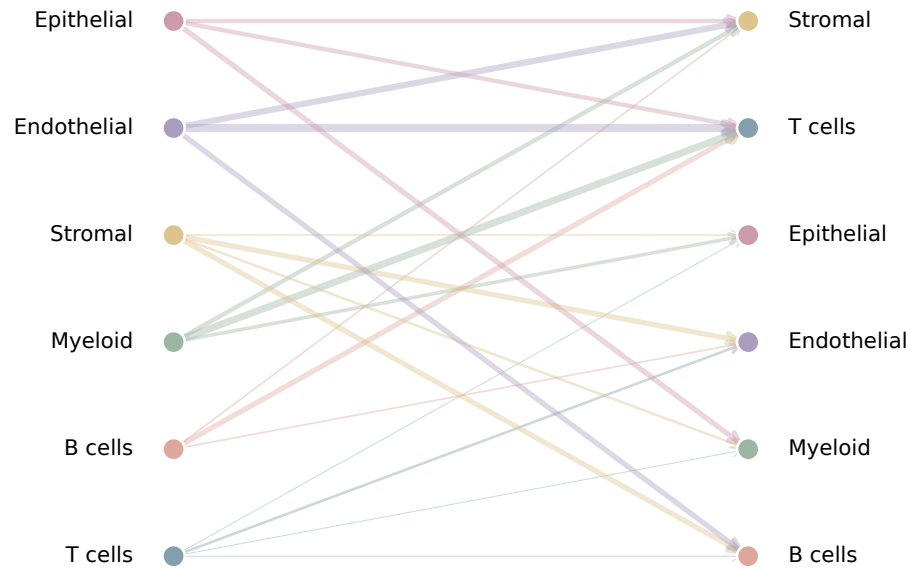
SYNTHETIC EXAMPLE

Lineage programmes over time



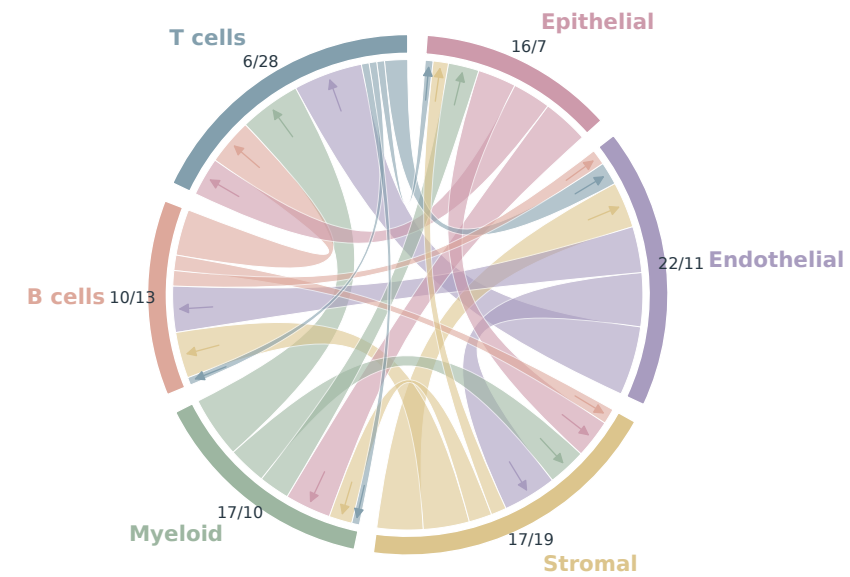
SYNTHETIC EXAMPLE

Candidate sender-receiver communication



SYNTHETIC EXAMPLE

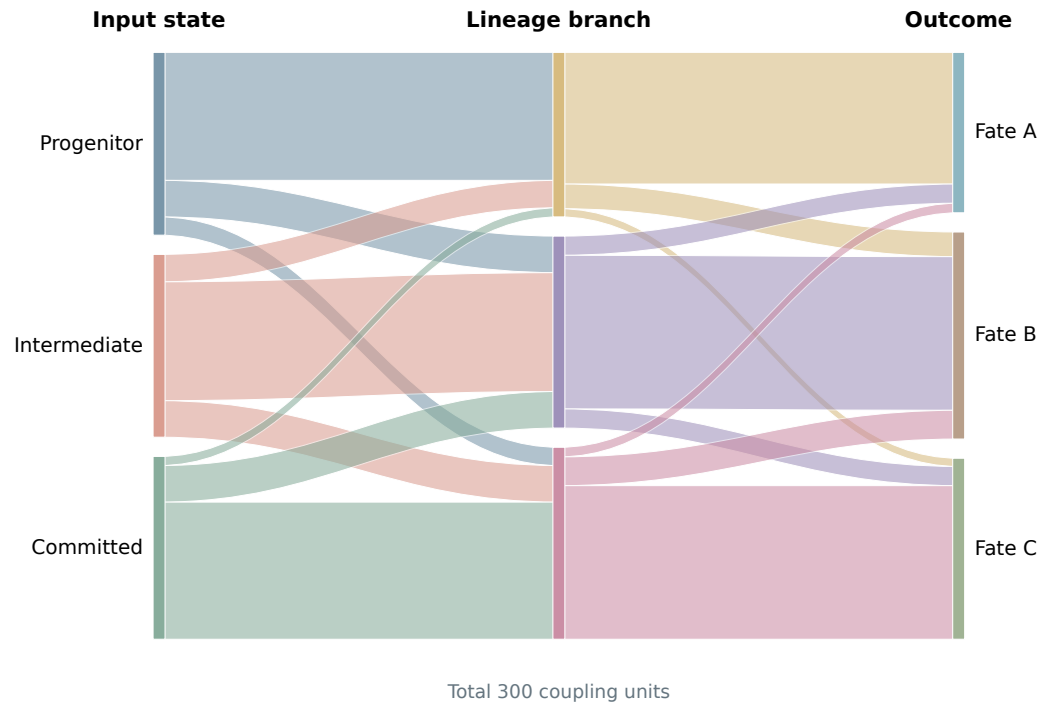
Candidate sender-receiver communication



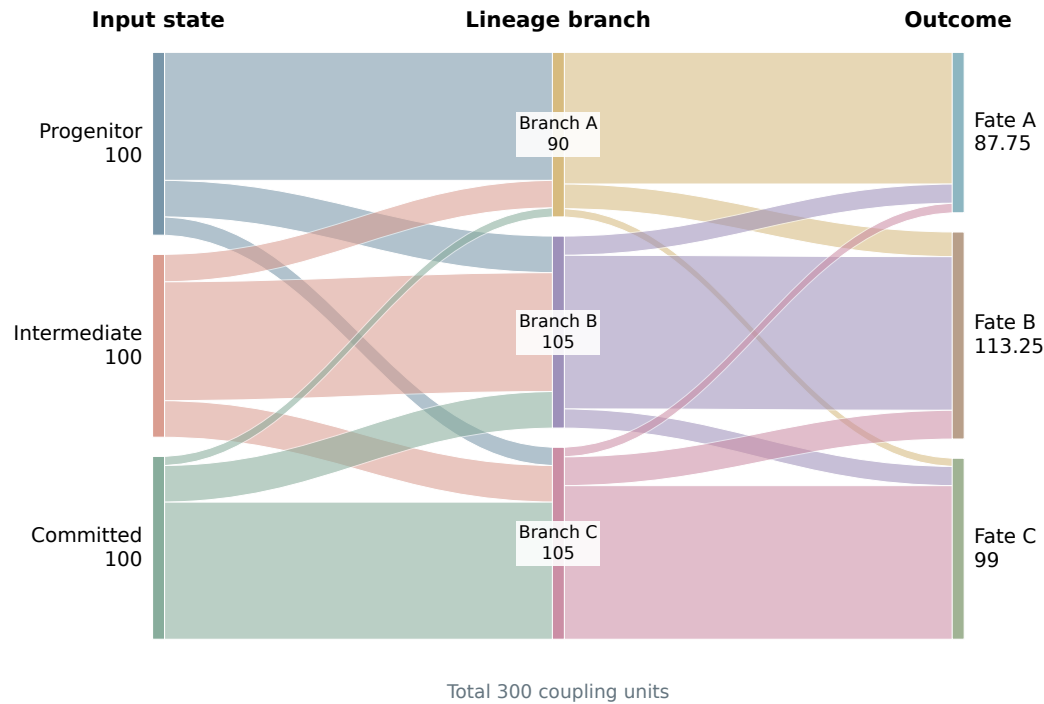
Node values: out / in (score); arrows mark receivers

SYNTHETIC EXAMPLE

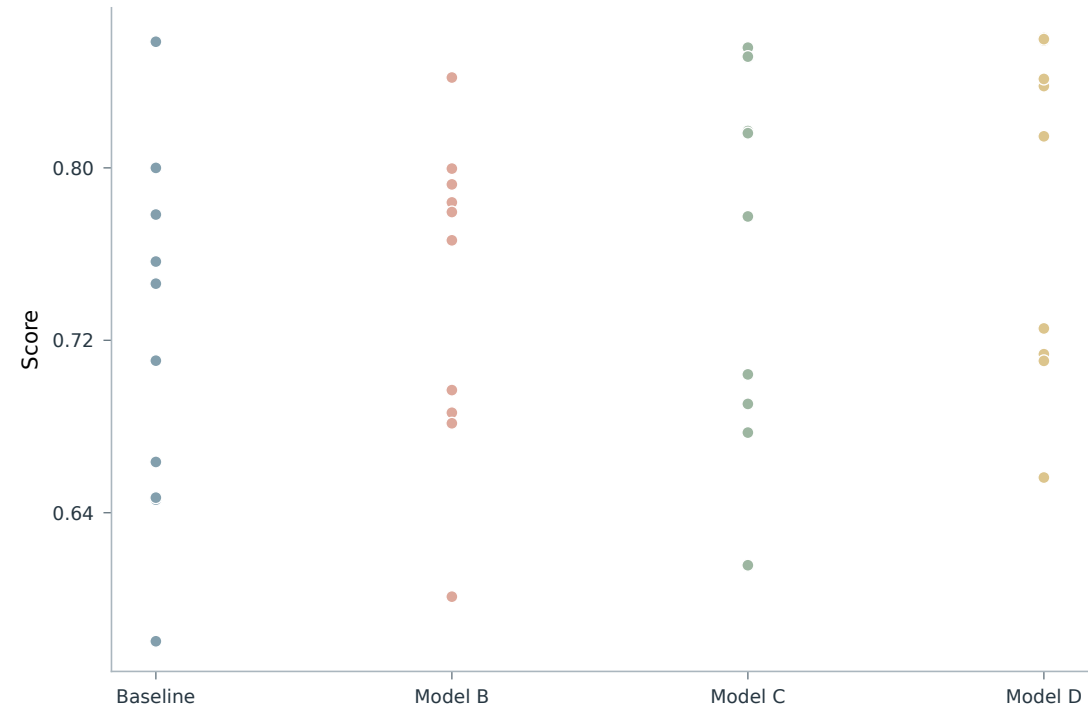
State transitions and conserved mass



State transitions and conserved mass



Matched-dataset method performance



SYNTHETIC EXAMPLE

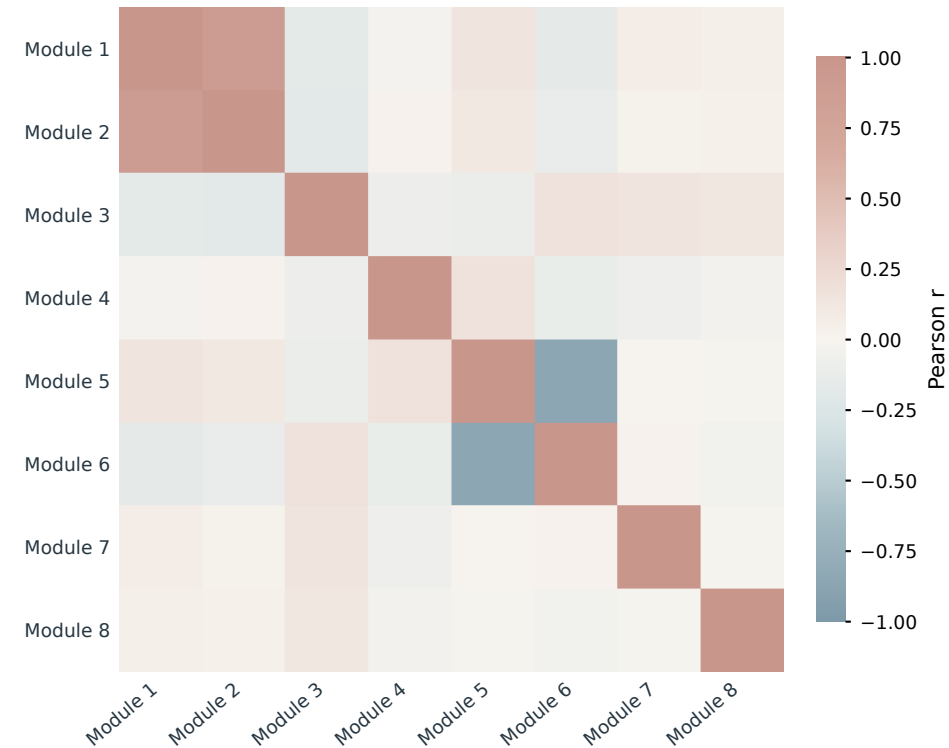
Matched-dataset method performance



Cell text = raw score; top = median paired difference

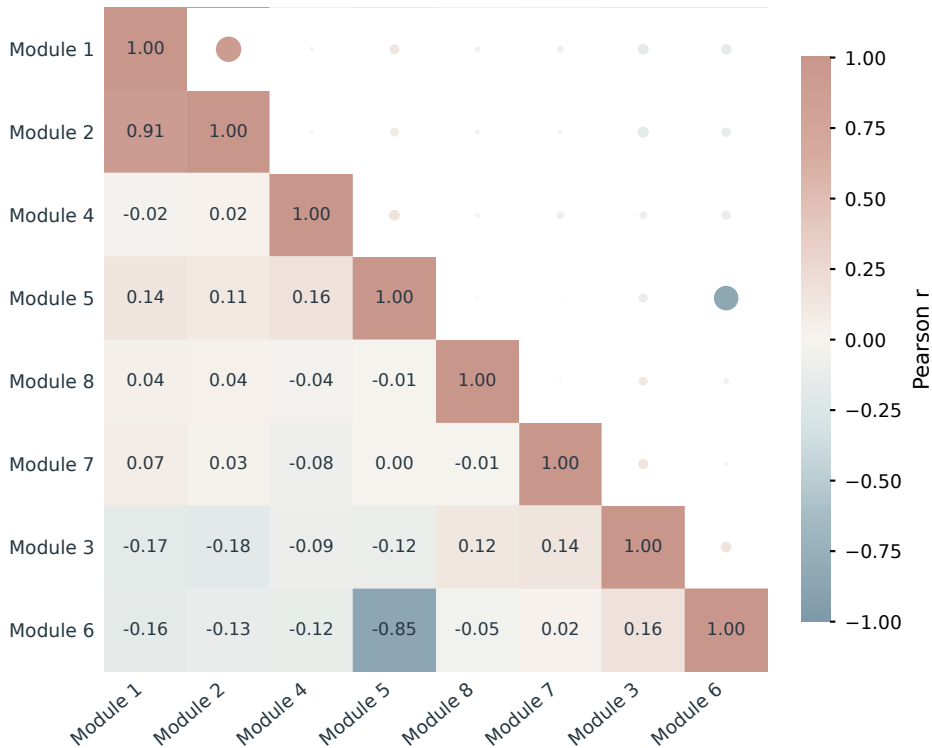
SYNTHETIC EXAMPLE

Correlation structure



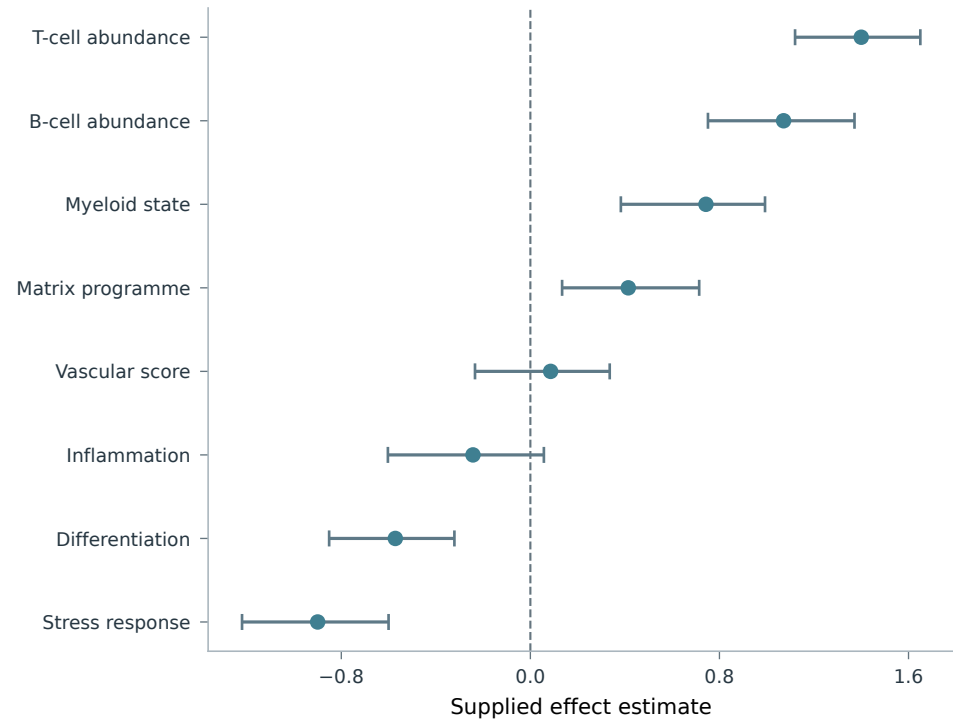
SYNTHETIC EXAMPLE

Correlation structure

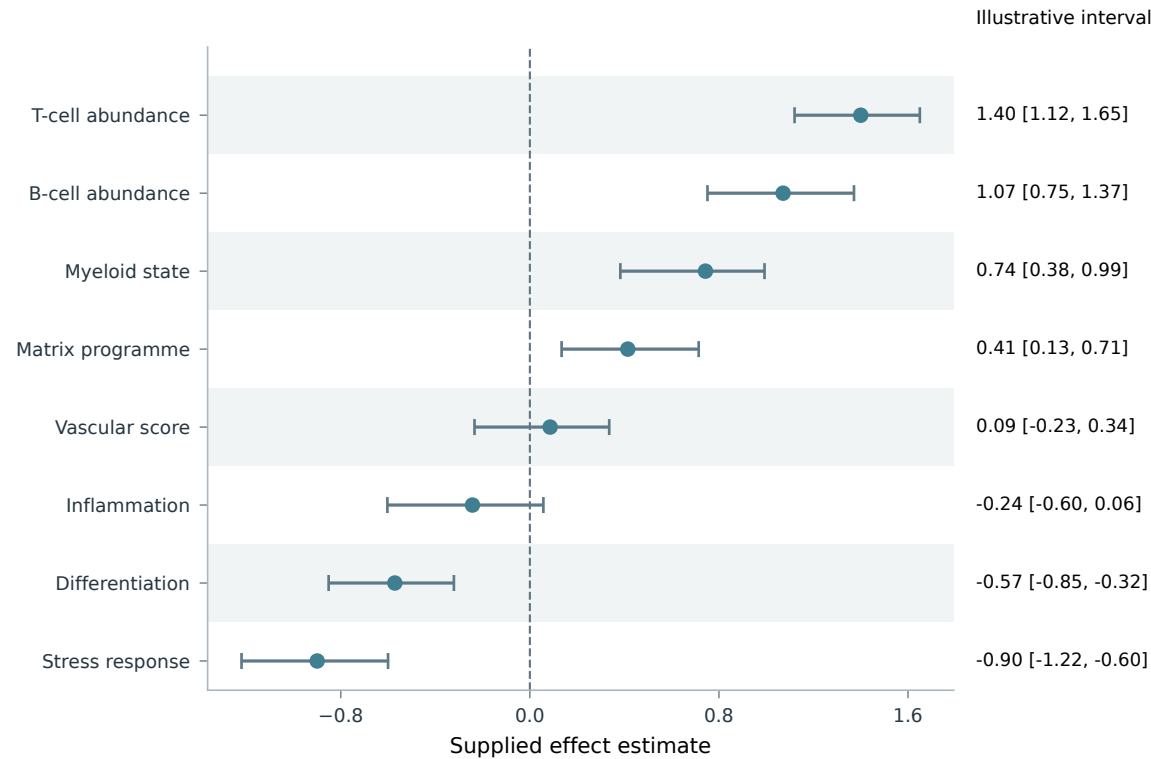


SYNTHETIC EXAMPLE

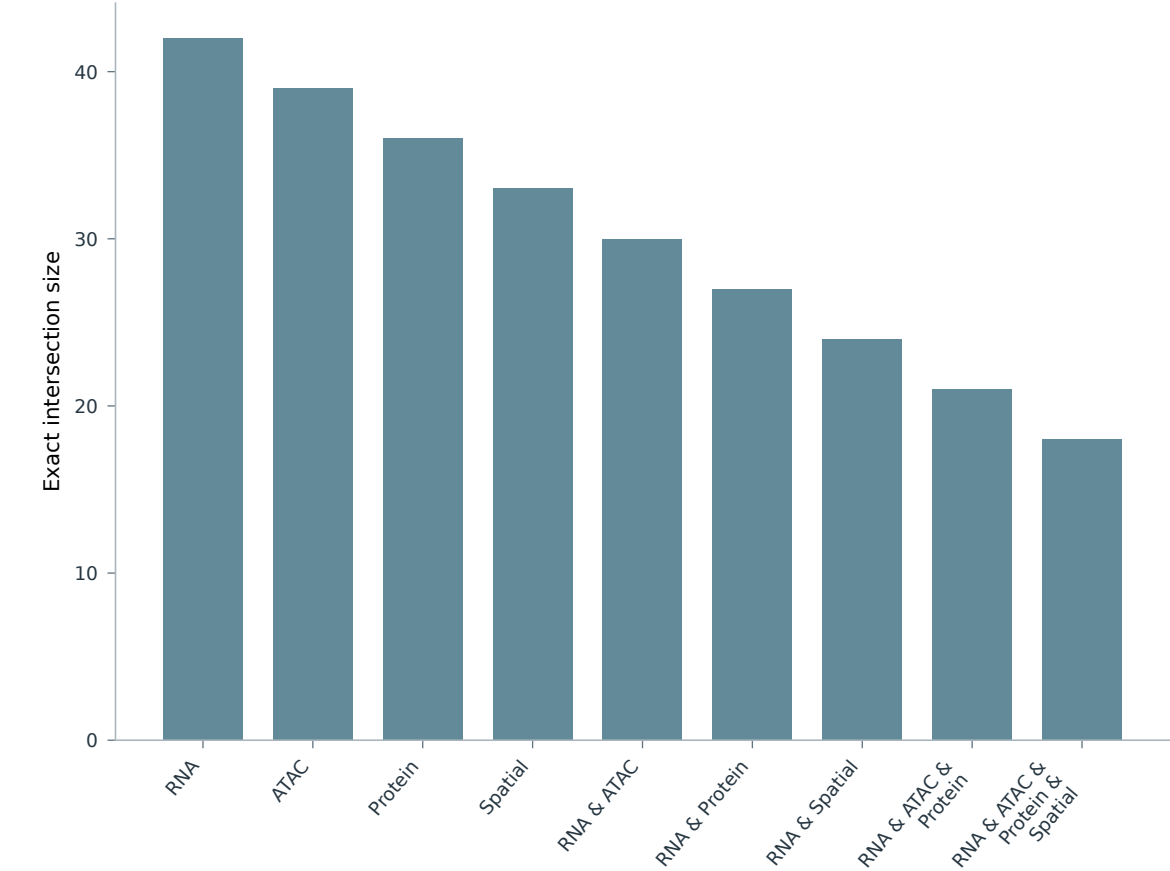
Effect sizes and uncertainty



Effect sizes and uncertainty

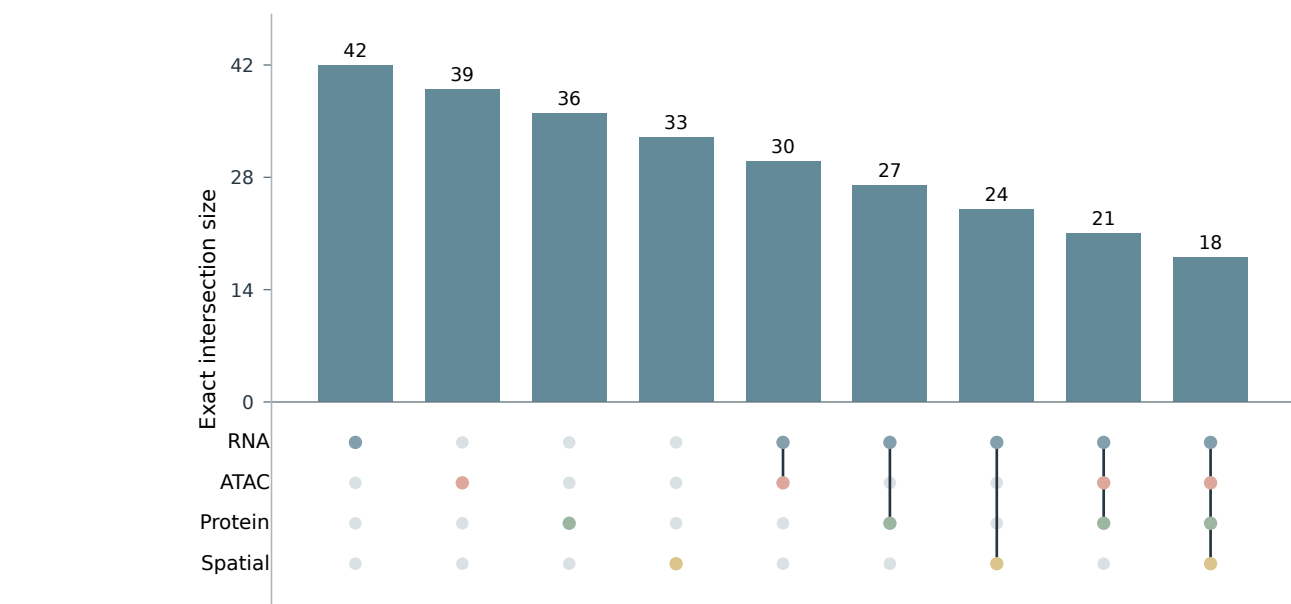


Exact intersections between gene sets



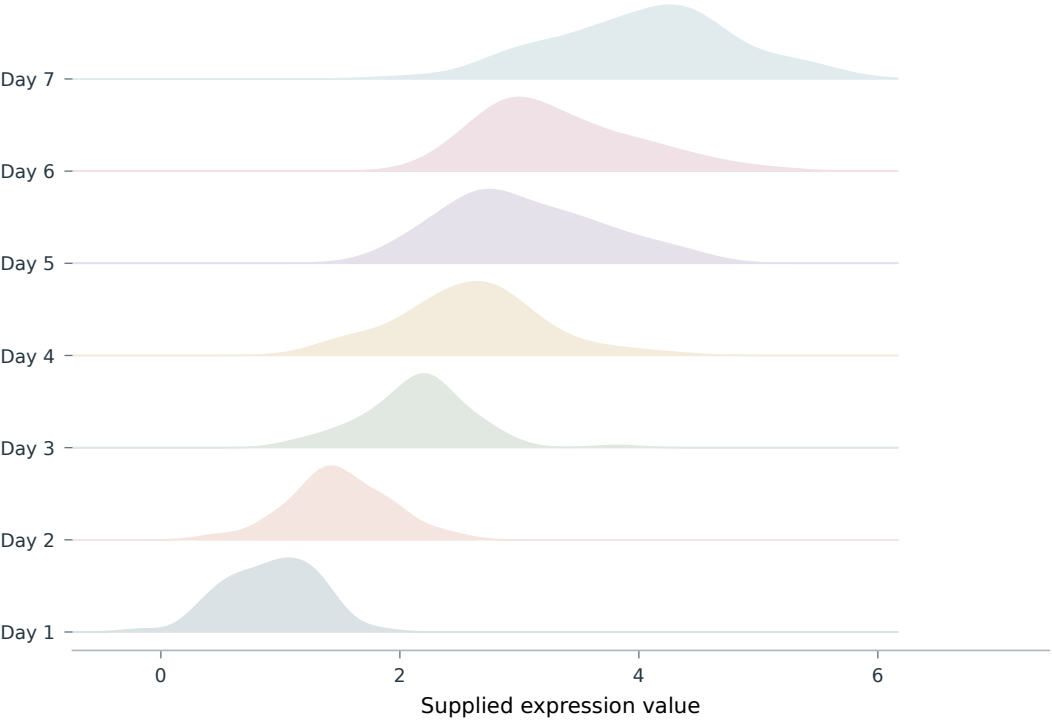
SYNTHETIC EXAMPLE

Exact intersections between gene sets



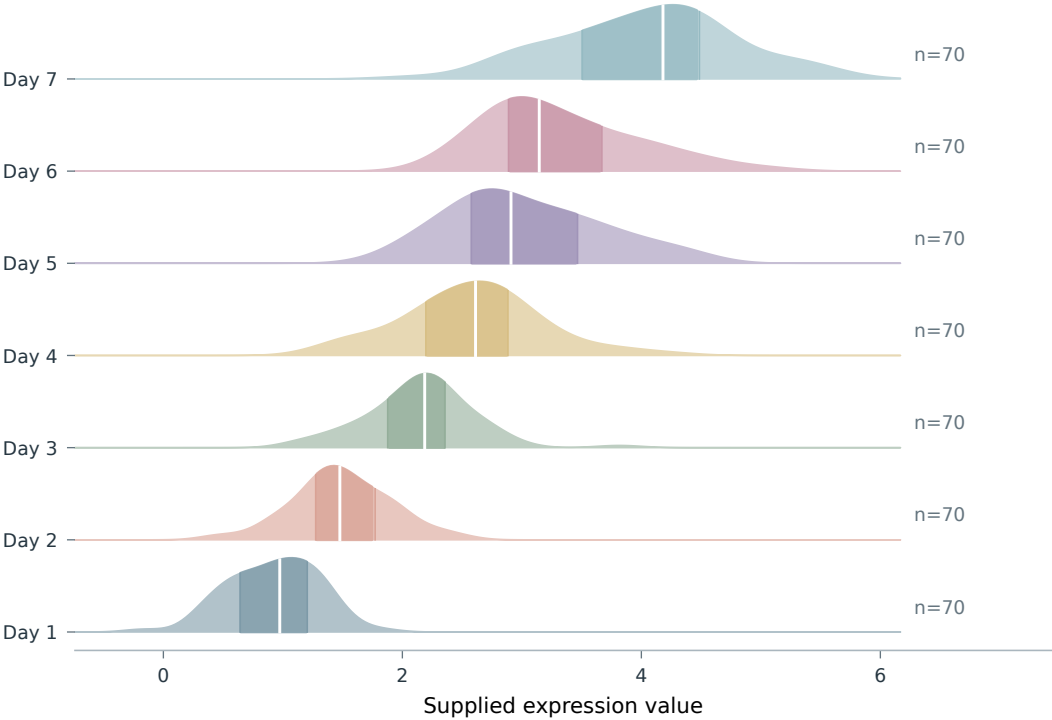
SYNTHETIC EXAMPLE

Expression distributions across ordered states



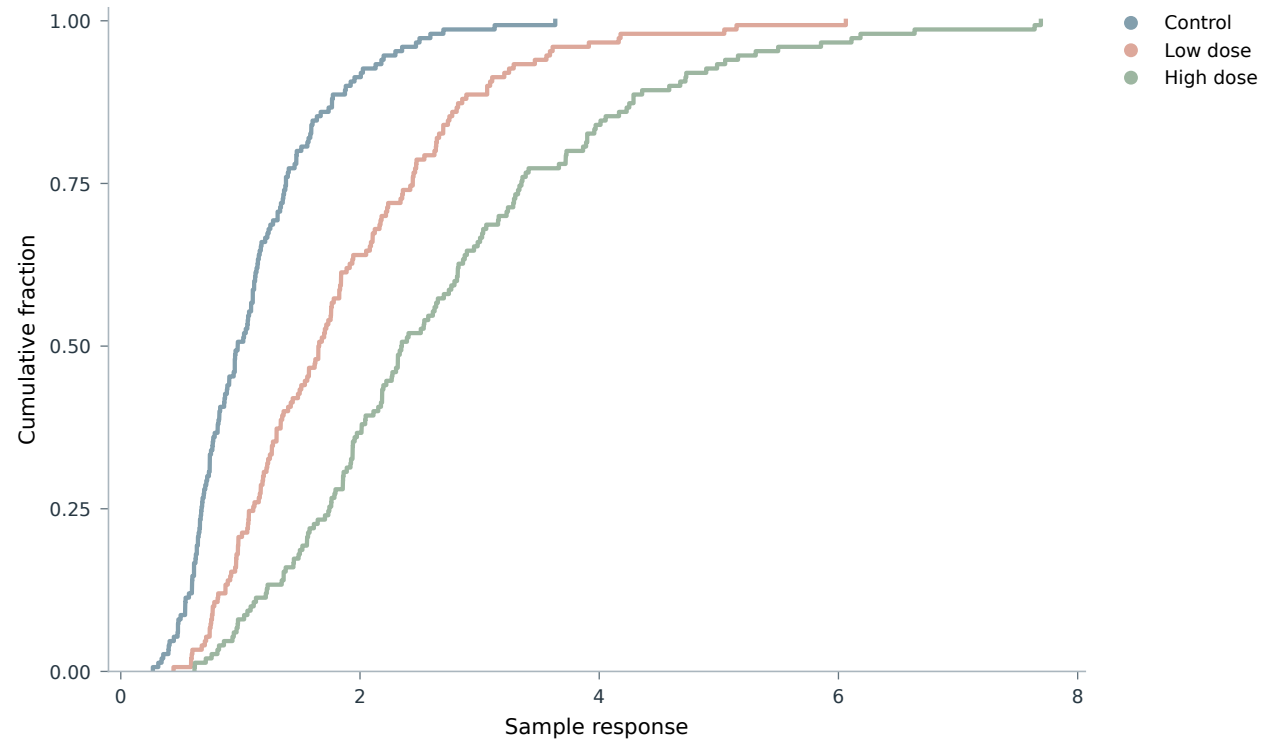
SYNTHETIC EXAMPLE

Expression distributions across ordered states



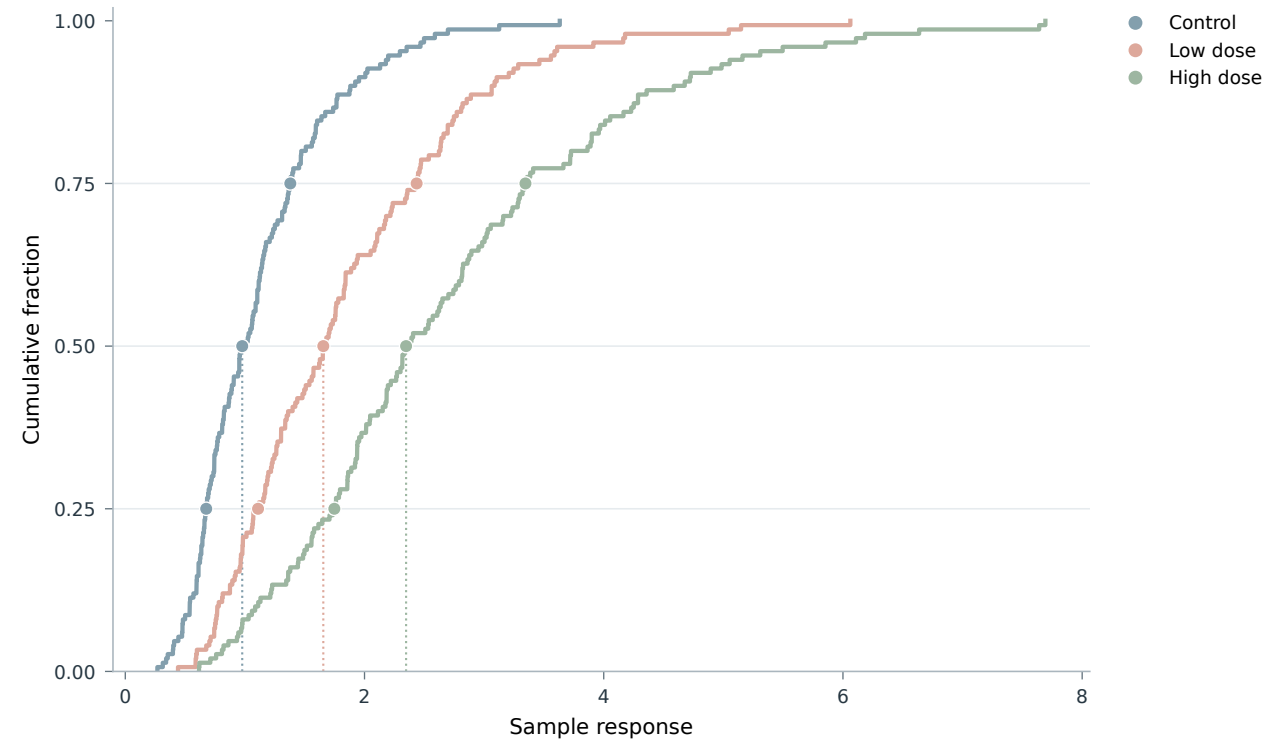
SYNTHETIC EXAMPLE

Cumulative sample distributions



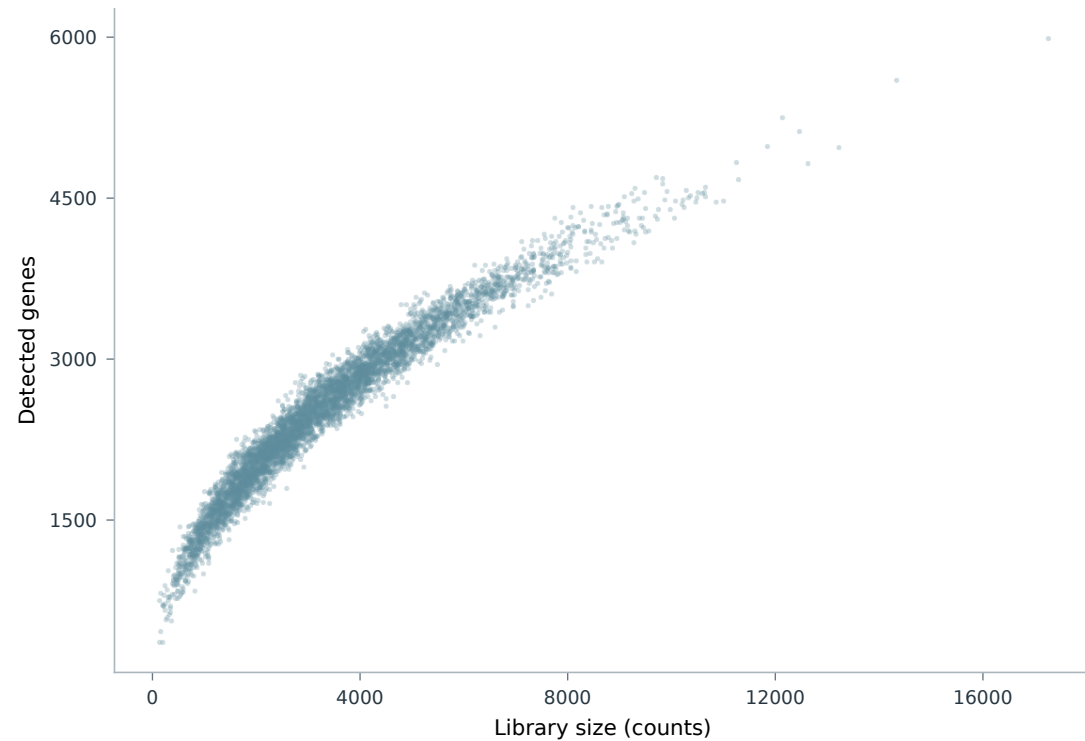
SYNTHETIC EXAMPLE

Cumulative sample distributions



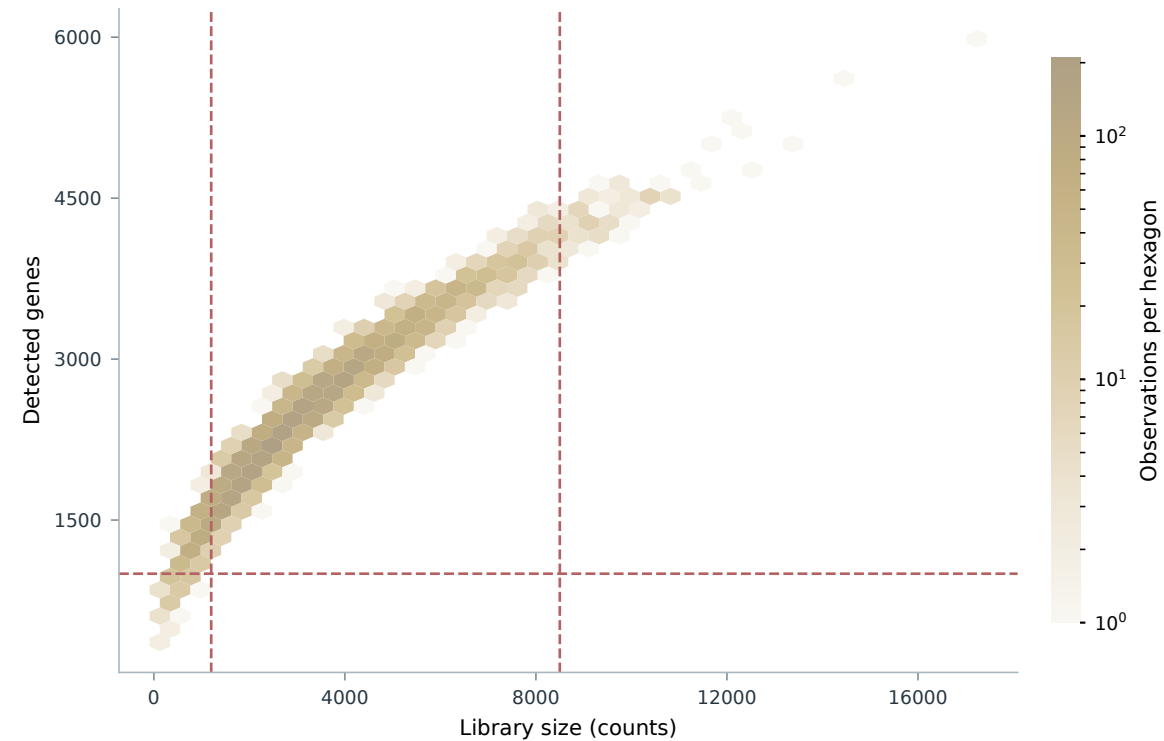
SYNTHETIC EXAMPLE

Density and quality-control thresholds



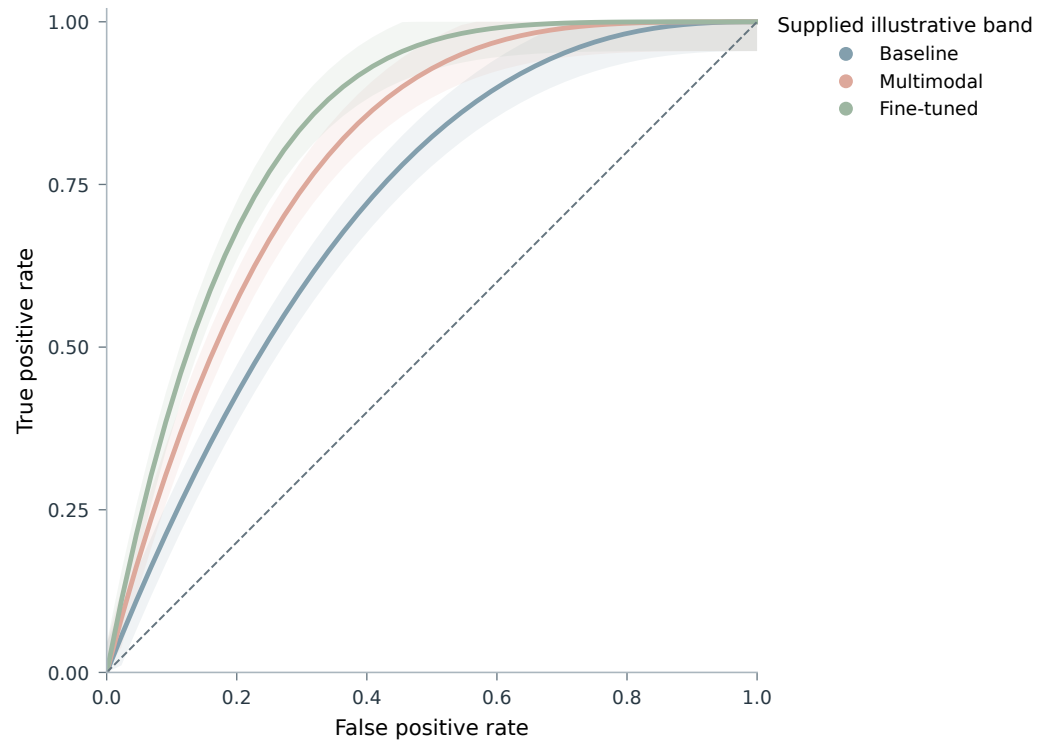
SYNTHETIC EXAMPLE

Density and quality-control thresholds



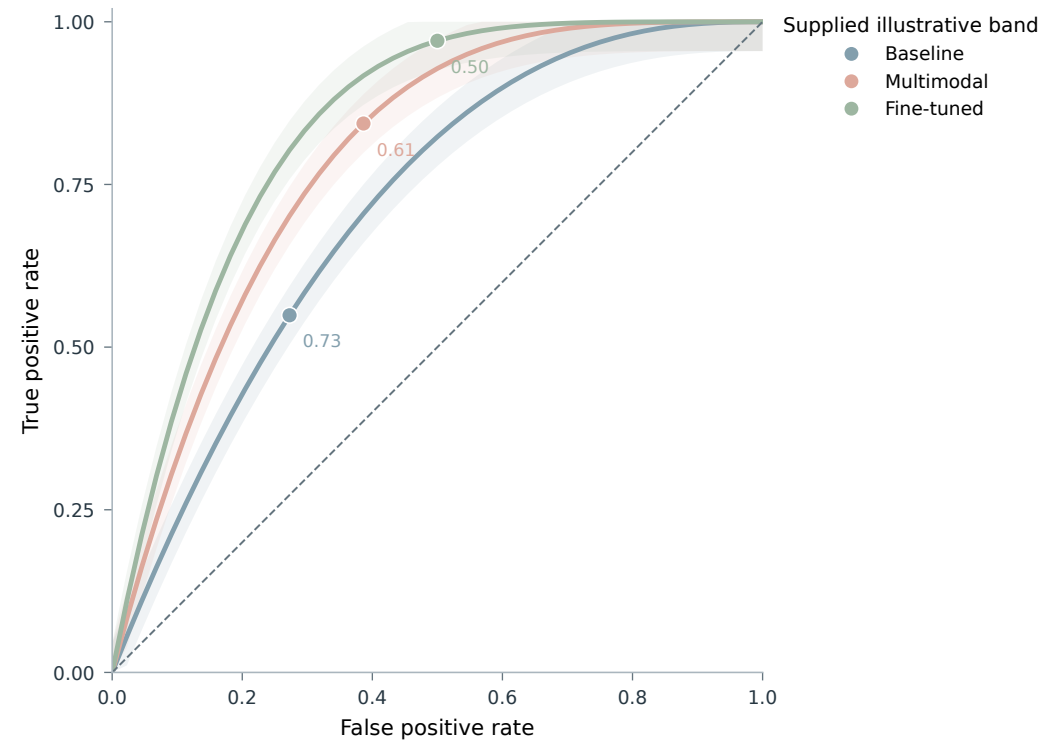
SYNTHETIC EXAMPLE

Discrimination across models



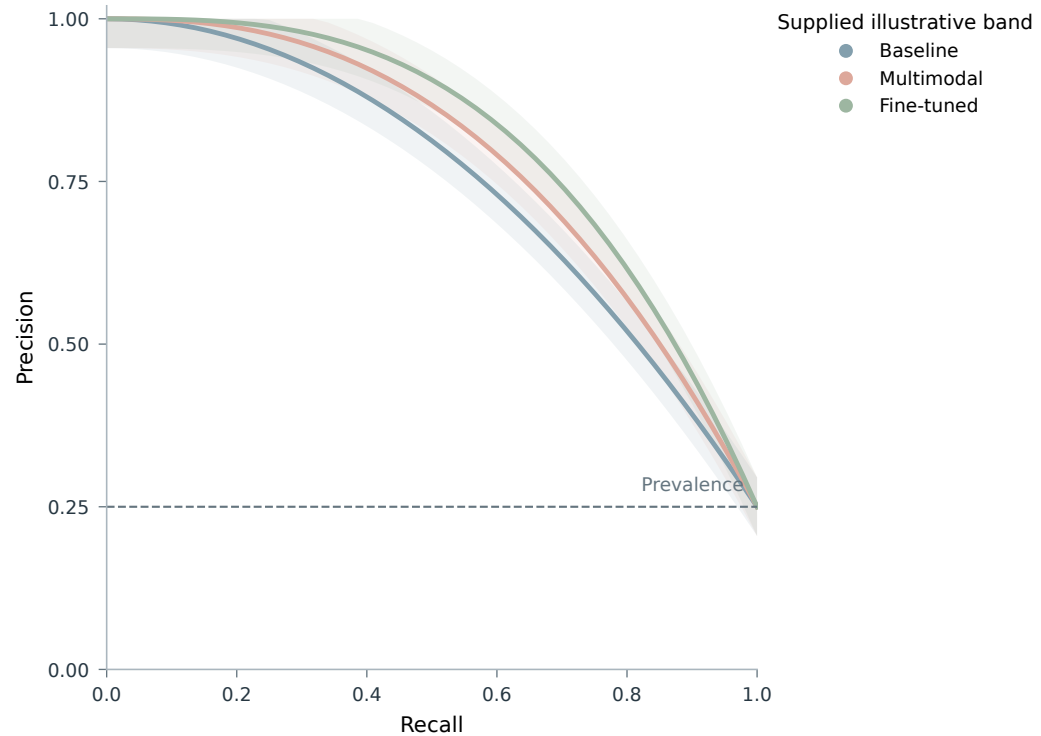
SYNTHETIC EXAMPLE

Discrimination across models



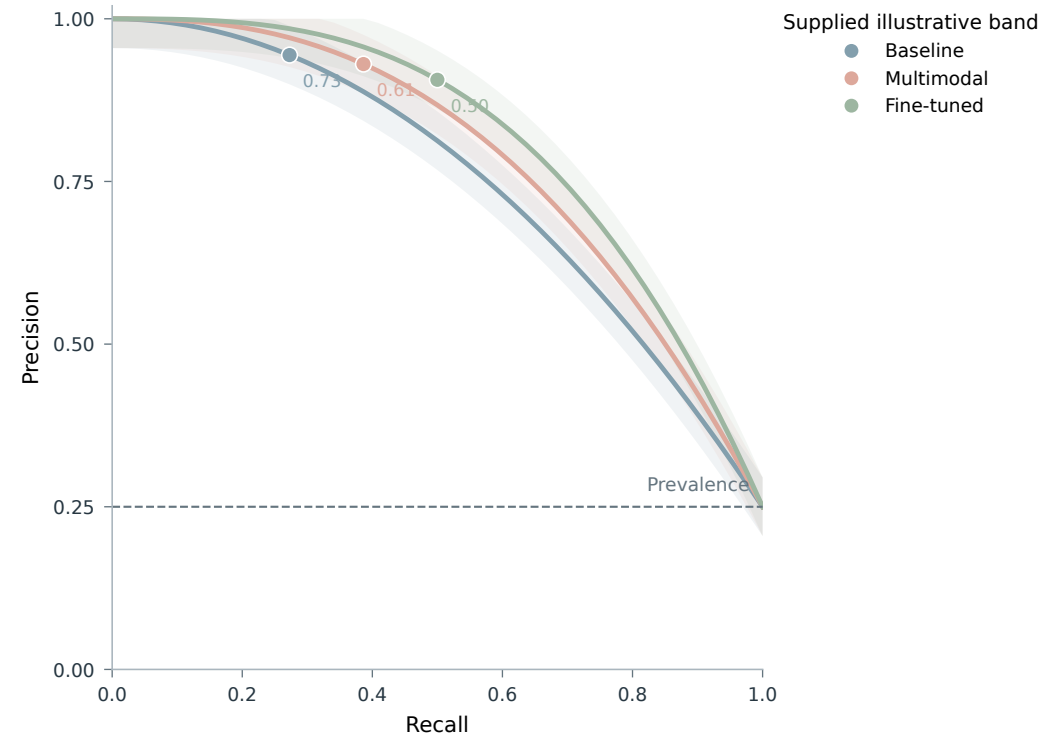
SYNTHETIC EXAMPLE

Precision-recall trade-offs



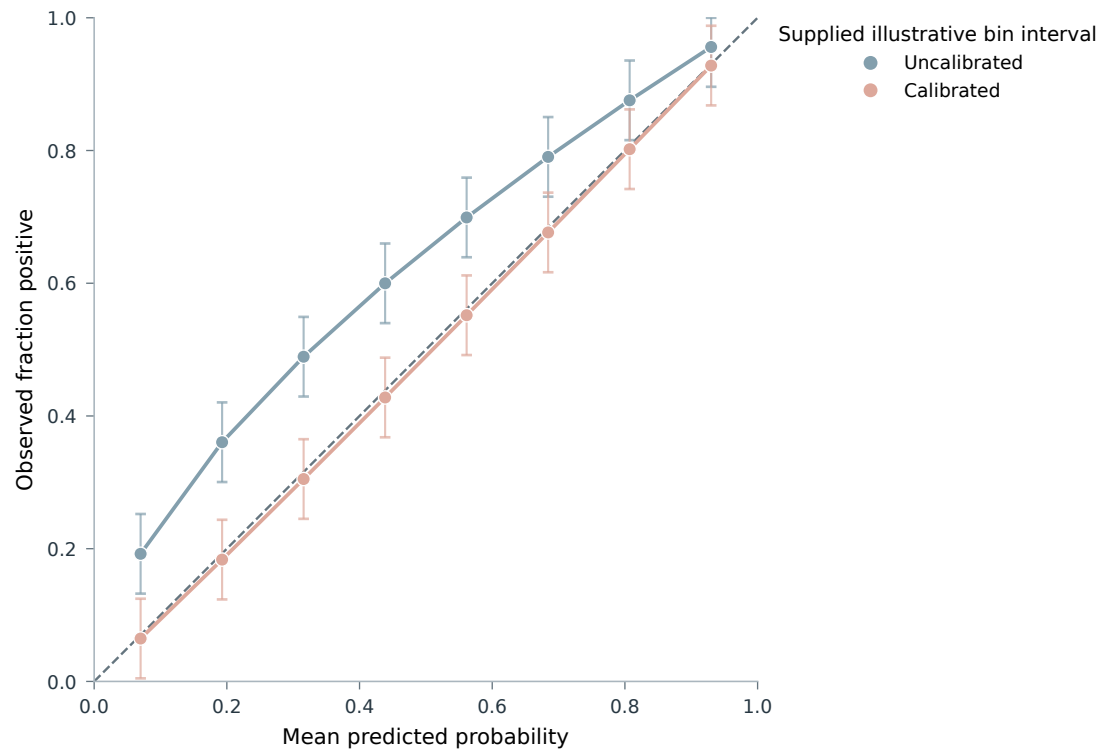
SYNTHETIC EXAMPLE

Precision-recall trade-offs



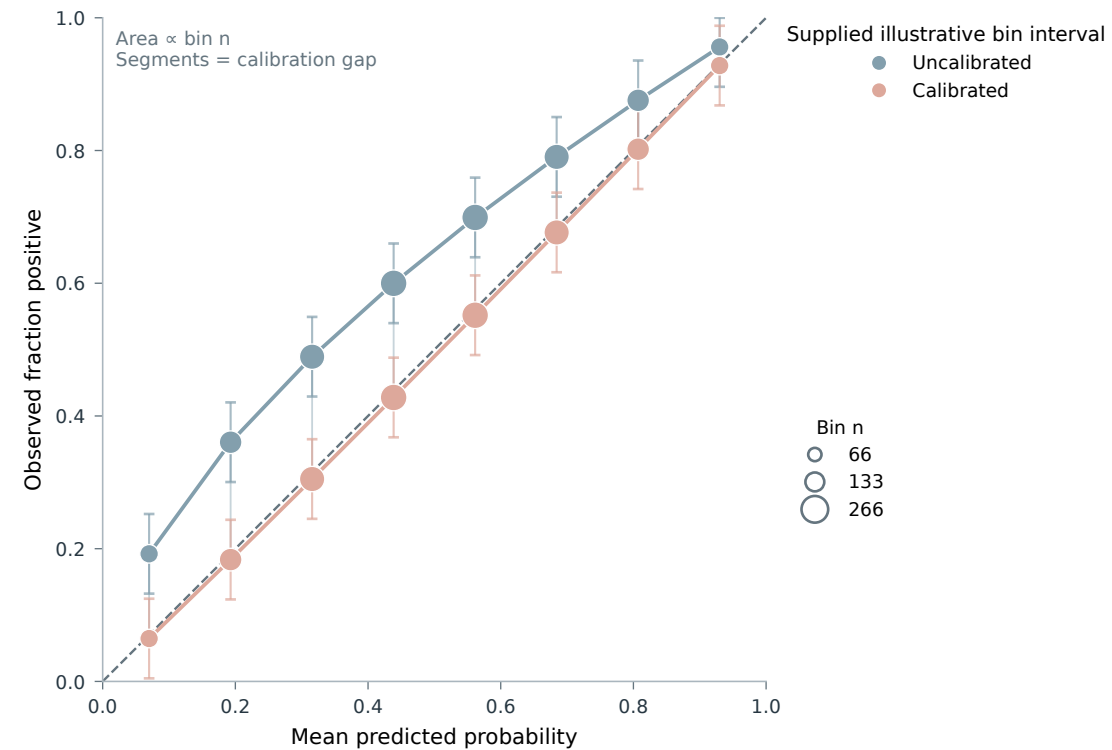
SYNTHETIC EXAMPLE

Probability calibration



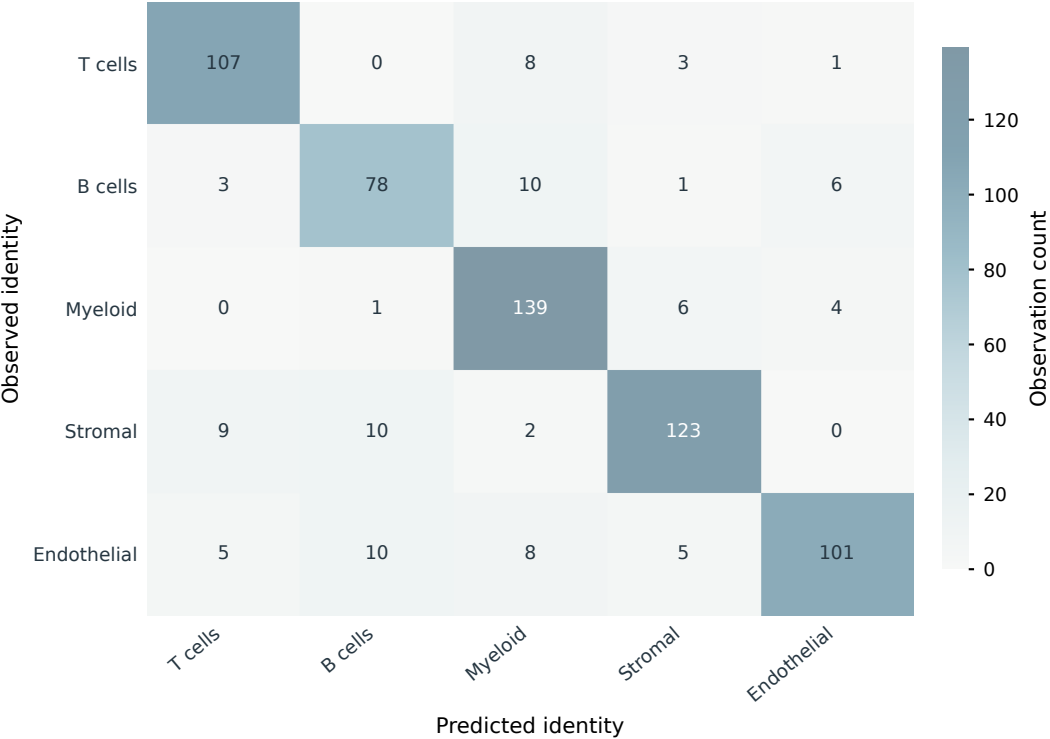
SYNTHETIC EXAMPLE

Probability calibration

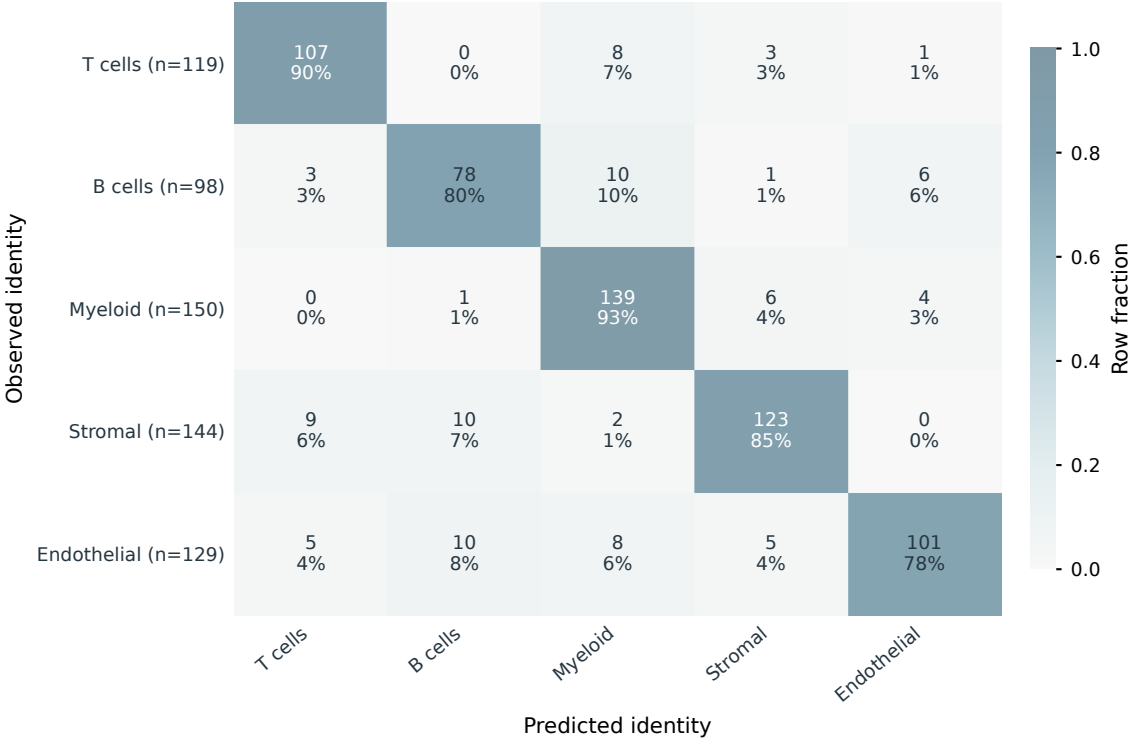


SYNTHETIC EXAMPLE

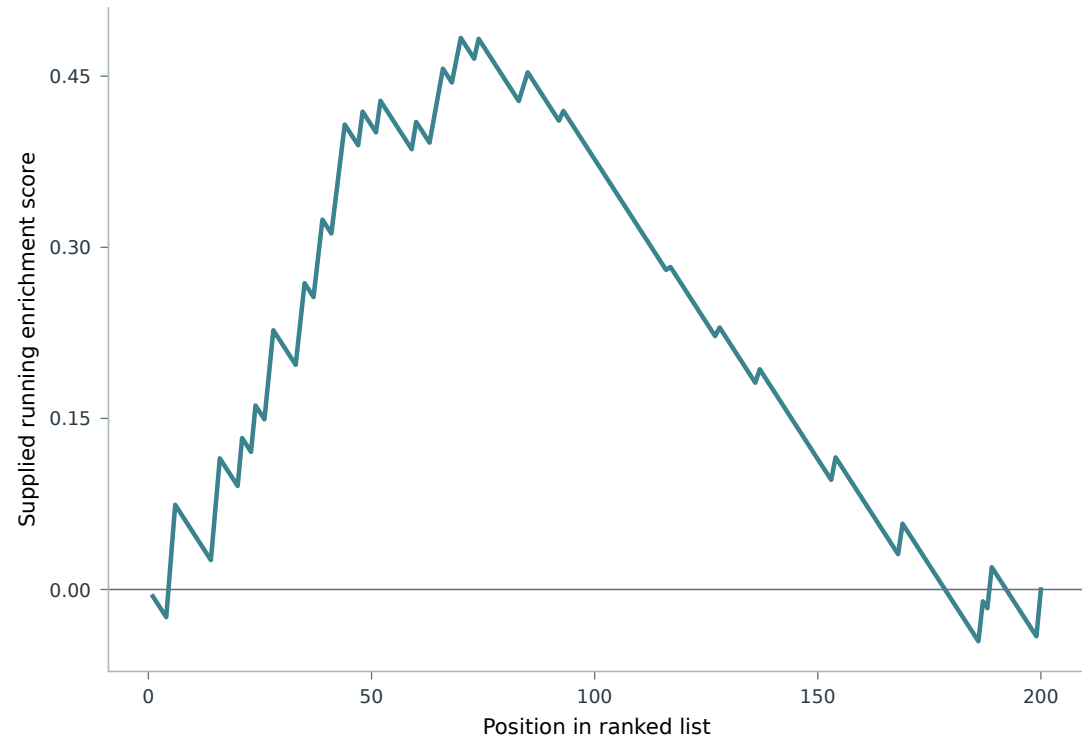
Classification errors by cell type



Classification errors by cell type

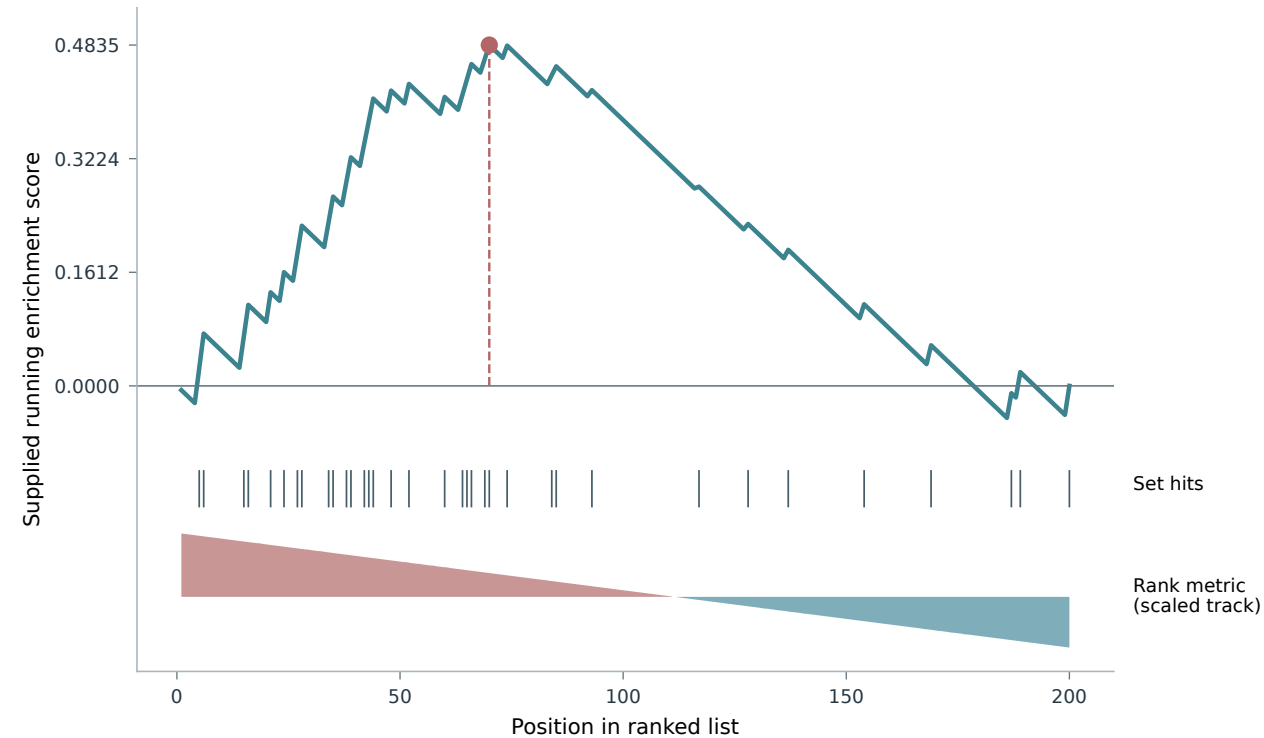


Ranked enrichment profile



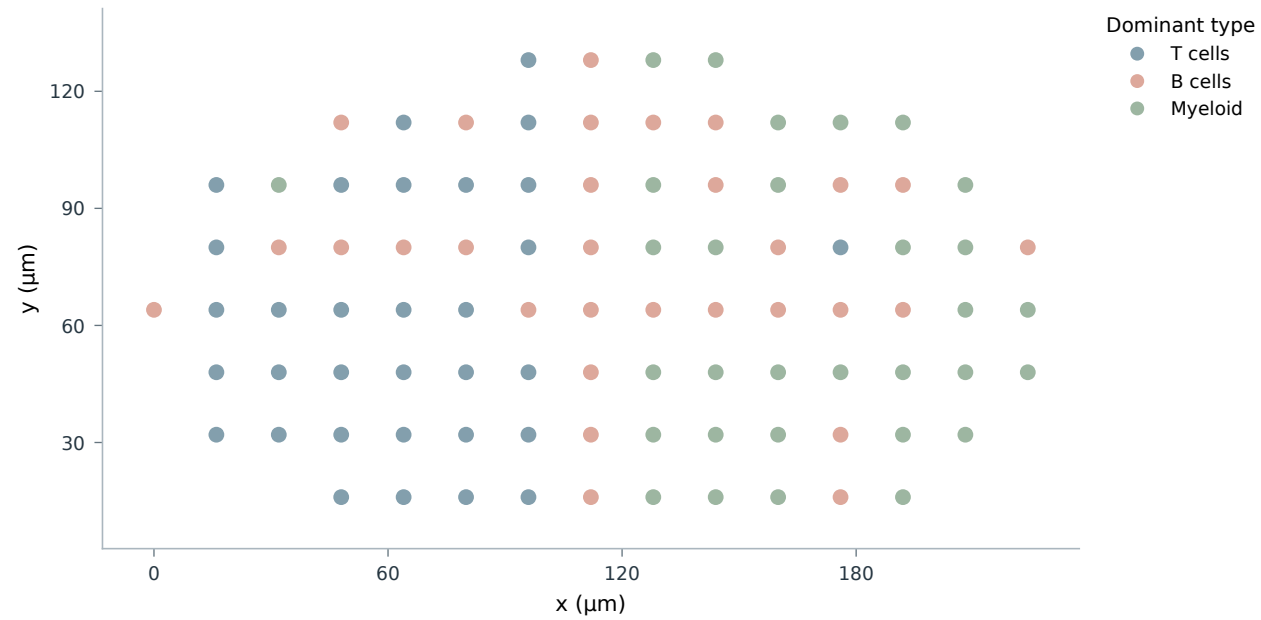
SYNTHETIC EXAMPLE

Ranked enrichment profile



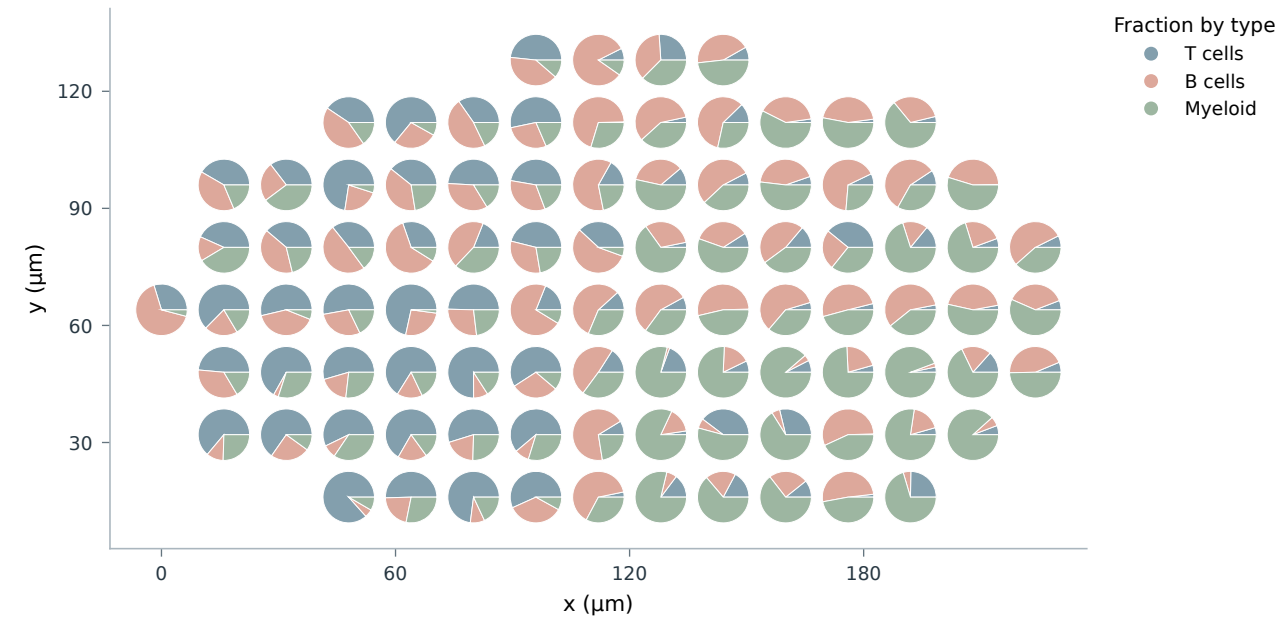
SYNTHETIC EXAMPLE

Spatial mixtures of cell identities



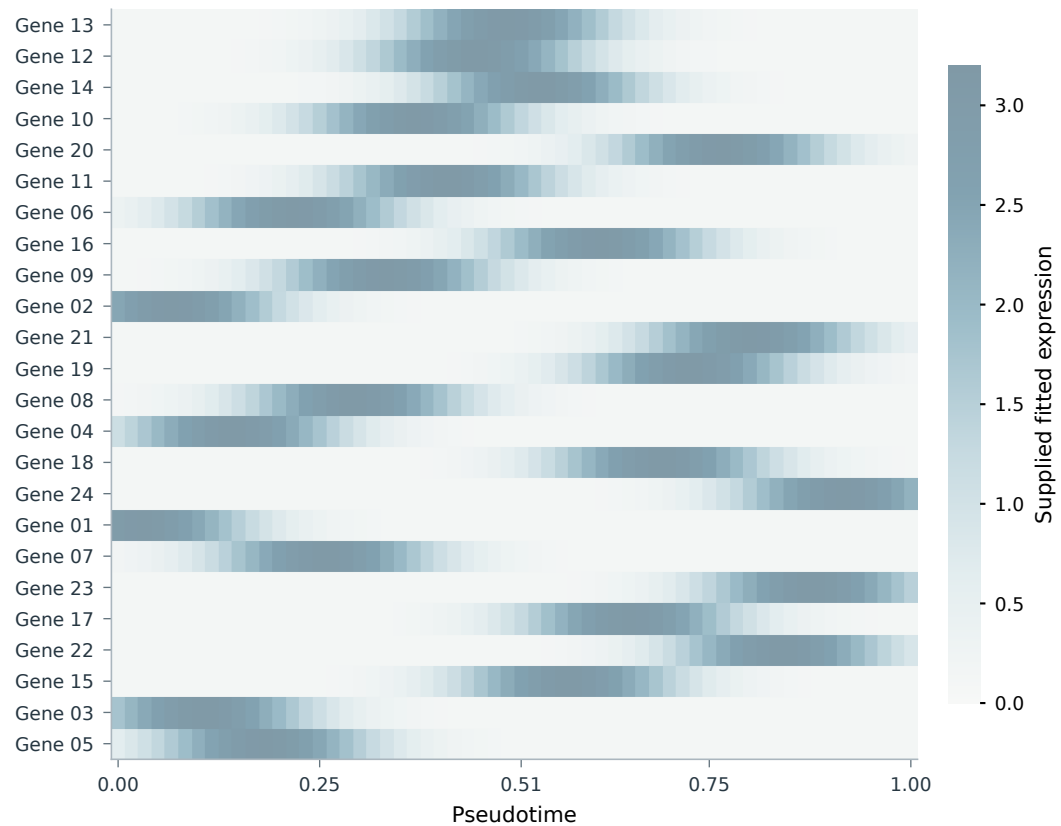
SYNTHETIC EXAMPLE

Spatial mixtures of cell identities



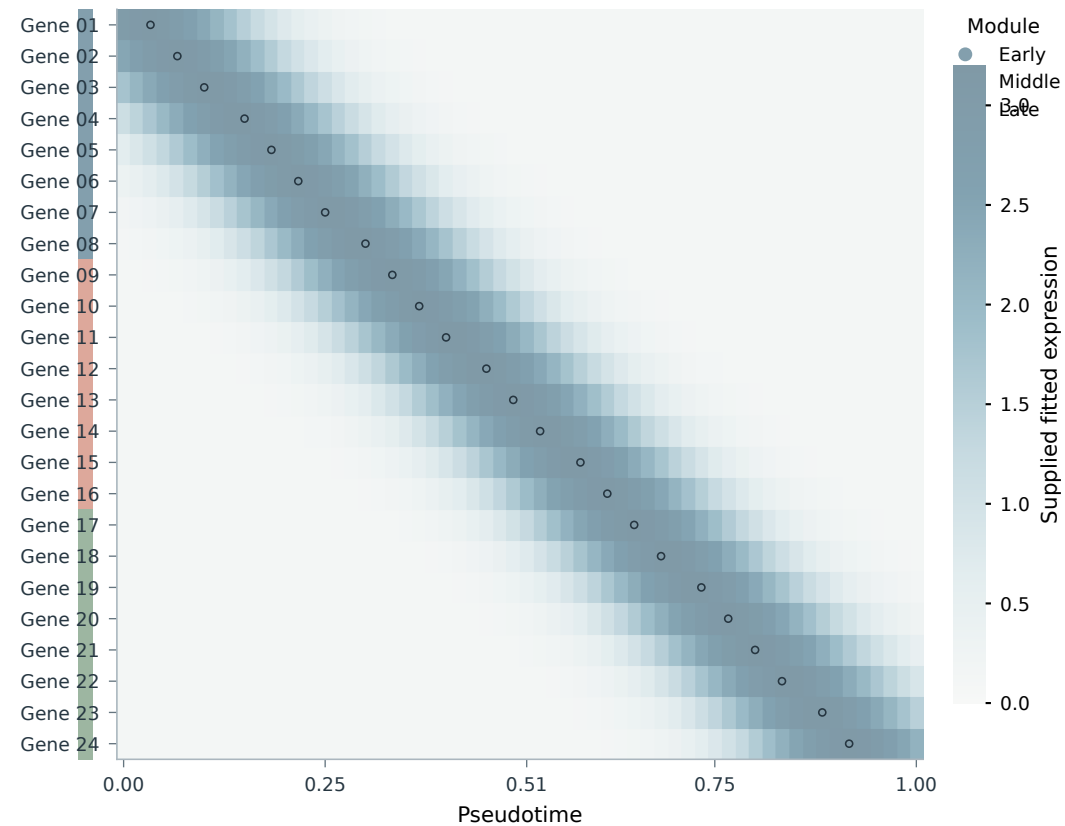
SYNTHETIC EXAMPLE

Temporal gene programmes



SYNTHETIC EXAMPLE

Temporal gene programmes



SYNTHETIC EXAMPLE